



MINISTRY OF INVESTMENT,
INDUSTRY AND TRADE
OF THE REPUBLIC OF UZBEKISTAN



FOREIGN INVESTORS COUNCIL
UNDER THE PRESIDENT OF
THE REPUBLIC OF UZBEKISTAN

Coca-Cola

ESG REVIEW REPORT

Closing the Loop: The Case for Circularity in a Linear World

Uzbekistan

JUNE • 2026

Table of *Contents*

6 Introduction

10 Water

- 11 - Closing the water loop
- 12 - Uzbekistan: water under pressure
- 13 - “Four National Taps”-Water Self-Sufficiency
- 14 - Water Reuse Systems by CCI

16 Waste

- 17 - From waste to resource
- 18 - Uzbekistan: significant challenges, decisive action
- 19 - Turning Trash into Treasure

20 Energy Transition

- 21 - A circular energy architecture
- 22 - Uzbekistan: significant challenges, decisive action
- 23 - The Carbon Neutral Train of Central Asia

24 Country Comparisons

- 25 - Belgium
- 28 - United Kingdom
- 31 - Romania
- 34 - Kazakhstan

38 Roadmap & Transition Design

- 39 - Governance
- 40 - Objectives
- 41 - Water Roadmap
- 45 - Waste Roadmap
- 50 - Energy Roadmap
- 54 - Appendix

EXECUTIVE BRIEF

Uzbekistan’s moment is *now*.

Uzbekistan’s moment is now. On the course to become one of Central Asia’s economic powerhouses, and its relative economic immaturity is a strategic advantage, Uzbekistan has the potential to enable a faster transition to a circular economy than more rigid, legacy-burdened peers. Paired with the right legislation and private-sector partnership, Uzbekistan can establish Central Asian leadership in circularity while simultaneously protecting the resource base upon which its growth depends.

Transitioning from a linear to a circular economy requires a combination of, among others, innovative mindsets, R&D, a supportive government, and a legislative foundation on which private technology providers, international financiers, and state utilities can rely.

Water, waste, and energy are not separate policy files, but rather interdependent systems. Efficient water use reduces energy consumption; waste-to-energy reduces landfill pressure and diversifies the power mix; cleaner energy reduces the carbon intensity of water treatment. A national circular economy strategy should integrate these systems under a single legislative and governance architecture, uniting the Office of the President, line ministries, international investors, and civil society around shared targets, transparent performance reporting, and sequenced investment priorities.

01- WATER:

Uzbekistan uses over 90% of its water on agriculture, with water stress more than six times the global benchmark. Structural reform through scarcity-indexed water pricing, zero liquid discharge mandates for industry, and PPP-financed infrastructure will be the primary levers for reversing this trajectory.

\$147bn

Nominal GDP (2025)

02- WASTE:

A recycling rate of ~5% against a global benchmark of 21%, with most waste ending up in substandard landfills. Introducing a national Extended Producer Responsibility scheme, circular procurement mandates, and investment in sorting and recycling infrastructure will be essential to closing this gap.

6-7%

Annual growth target

03- ENERGY:

Renewable energy accounts for just ~1% of the mix, making it one of the most fossil-intensive economies in the region. A Contracts for Difference auction mechanism, a mandatory carbon registry, and a phased Emissions Trading System will be the critical enablers for scaling renewable.

\$240bn

GDP goal by 2030

About this report

This report is prepared by Deloitte Norway, Deloitte Uzbekistan, with the support of Foreign Investors Council and its member The Coca-Cola Company. The report is designed to support an evidence-informed scoping that maps global circular economy trends and establishes Uzbekistan's current baseline from legislative, industrial, and competitive perspectives.

Its primary audience are Uzbek decision makers, hereunder the Office of the President, the Cabinet of Ministers, the National Committee for Environmental Protection, line ministries and regulators, international investors, development partners, business associations, and civil society. The report also aims to strengthen communication of, and commitment to, circularity and its beneficial effects on the society at large, ensuring that Uzbekistan is informed and engaged about the circular transition, ultimately benefiting from a society valuing productivity, resource efficiency, and intelligent use of scarce resources.

The report integrates best practices, legislative status, market research, case examples, and country benchmarking to provide insights to specific trends driving circular economy initiatives and relevant first-step actions to promote the transition from a linear to a circular economy in Uzbekistan.

PRIMARY AUDIENCE

- Office of the President
- Cabinet of Ministers
- National Committee for Environmental Protection
- Line ministries and regulators
- International investors and development partners
- Business associations and civil society

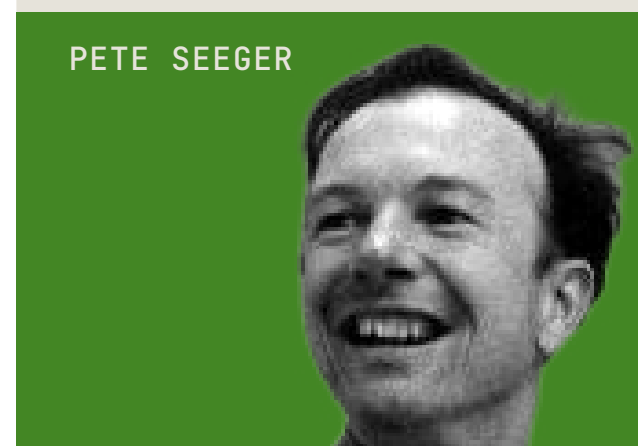
REPORT METHODOLOGY

The report integrates best practices, legislative status, market research, case examples, and country benchmarking to provide insights to specific trends driving circular economy initiatives and relevant first-step actions.

It also aims to strengthen communication of, and commitment to, circularity and its beneficial effects on society at large, ensuring that Uzbekistan is informed and engaged about the circular transition.

We throw away *billions* of dollars every day

PETE SEEGER



“If it can’t be reduced, reused, repaired, rebuilt, refurbished, refinished, resold, recycled, or composted, then it should be restricted, designed or removed from production.”



Introduction

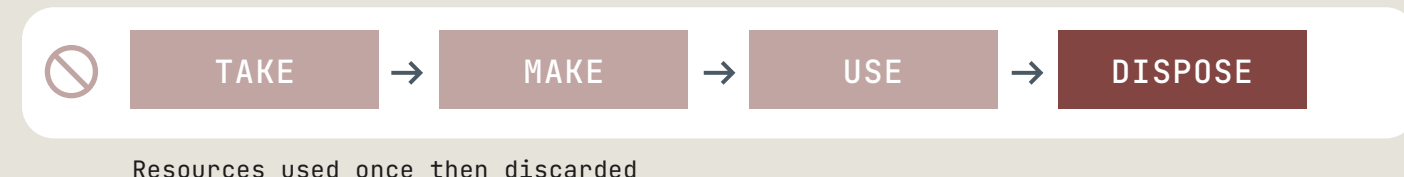
The food you leave on the plate is just the tip of the iceberg in a supply chain dealing with spoils at every stage.

Mountains of discarded clothes worn just once or twice create toxic, space-visible environmental hazards. Electronics are discarded long before their actual lifetime, where invaluable rare earth minerals and materials end up on landfills, never to be used again. These are just some of the consequences originating from a linear economy where resources are produced and consumed without thinking about resource loss or their afterlife.

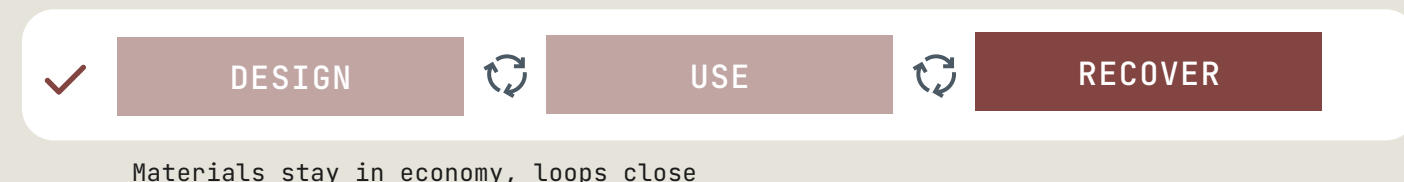
Unlike the traditional linear model of “take, make, dispose,” the circular economy focuses on designing out waste and pollution, keeping products and materials in use for as long as possible, and regenerating natural systems. By adopting circular principles, countries can reduce their dependence on finite resources, lower environmental impacts, and create more sustainable economic growth, thus enhancing national competitiveness and resilience.

Key areas such as water management, waste reduction, energy use, and CO₂ emissions illustrate the practical benefits of a circular economy. Together, these strategies not only support environmental sustainability but also strengthen a nation’s ability to adapt to economic and environmental shocks, thereby boosting long-term competitiveness and resilience.

LINEAR



CIRCULAR



KEY BENEFITS

- Reduce finite resource dependence
- Enhance national competitiveness
- Lower environmental impacts
- Build economic resilience

INTRODUCTION

Compared to more mature economies, where the transition to a circular economy can be challenging due to legacy systems and red-tape regulation, complex and rigid supply chains, and ingrained consumer behavior, Uzbekistan’s relative immaturity can be a competitive advantage and accommodate a faster transition towards a circular economy. The economy is experiencing strong growth with a nominal GDP of USD 147 billion in 2025, expecting growth of between 6%–7% annually, and aiming for, or exceeding, 240 billion USD by 2030, turning Uzbekistan into one of Central Asia’s power trains.

These growth ambitions should be accompanied by proactive and offensive regulations and policies aiming to promote productivity, resource efficiency, and intelligent use of scarce resources. Uzbekistan has the potential to take leadership within Central Asia concerning circular economy and enable sustainable growth, building national competitiveness and resilience along the way.

\$147bn

Nominal GDP (2025)

6-7%

Annual growth target

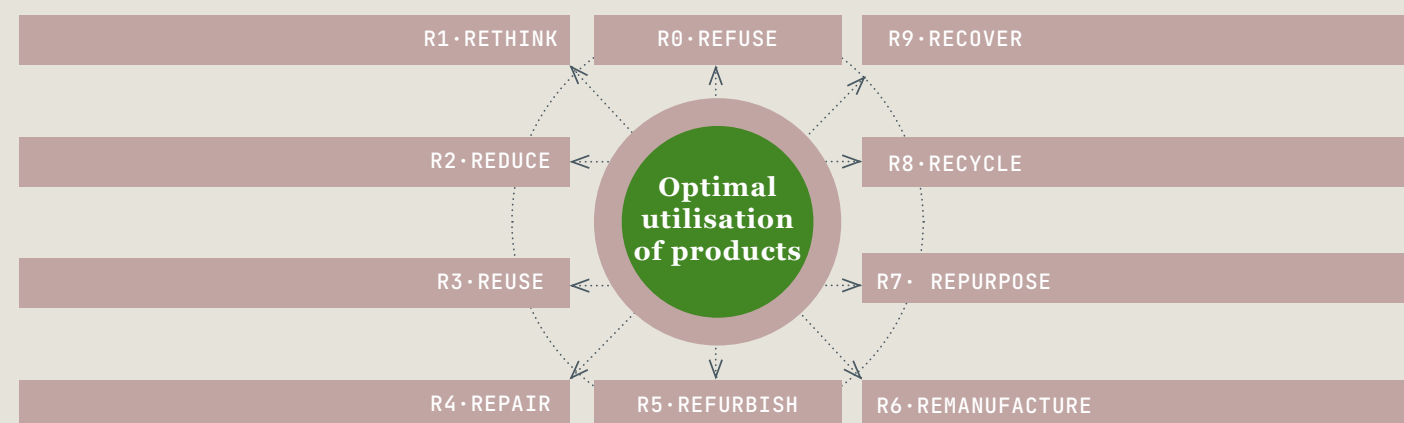
\$240bn

GDP goal by 2030

THE CIRCULAR ECONOMY FRAMEWORK

Nine *essential* strategies for circularity.

There are nine essential strategies for a successful transition from a linear to a circular economy. These strategies aim to reduce resource use and waste. The strategies are presented from the most efficient to the least efficient strategy. The figure also includes a strategy where products are abandoned altogether or exchanged by radically different products.



PREFERRED – AVOID RESOURCE USE

R0-REFUSE	Make product redundant by abandoning its function or by offering the same function with a radically different product	R5-REFURBISH	Restore an old product and bring it up to date
R1-RETHINK	Make product use more intensive (e.g. by sharing product)	R6-REMANUF	Use parts of discarded product in a new product with the same function
R2-REDUCE	Increase efficiency in product manufacture or use by consuming fewer natural resources and materials	R7-REPURPOSE	Use discarded product or its parts in a new product with a different function
R3-REUSE	Reuse by another consumer of discarded product which is still in good condition and fulfills its original function	R8-RECYCLE	Process materials to obtain the same (high grade) or lower (low grade) quality
R4-REPAIR	Repair and maintenance of defective product so it can be used with its original function	R9-RECOVER	Incineration of material with energy recovery – last resort

FIGURE 2

9Rs framework in a circular economy. Uzbekistan's opportunity: Early-stage economies can design circular principles into new infrastructure from the outset. The most efficient strategies (R0-R2) offer the greatest environmental and economic return.

CIRCULAR ECONOMY AND ITS APPLICATIONS

What is a *circular* economy?

Circular economy is broadly defined as a regenerative system designed to minimize resource input, waste, emissions, and energy leakage by slowing, closing, and narrowing material and energy loops. Unlike the traditional linear economy, which follows a “take-make-dispose” pattern of resource extraction and eventual waste, a circular model prioritizes longer-lasting design, maintenance, repair and remanufacturing. This transition extends beyond solid materials to encompass the entire resource flow of an economy, including the management of water and atmospheric carbon.

NO LONGER “NICE TO HAVE”

For modern industrial economies, adopting these principles is no longer just “nice to have” but a core strategy for national economic competitiveness. By reducing reliance on volatile global supply chains and imported raw materials countries can stabilize production costs and protect their economies from geopolitical shocks which are occurring on a more frequent basis. This shift is not easy. It is a task even highly industrialized and developed nations, with plenty of private capital and engineering competence, struggle to achieve.

The shift mandates high levels of investment in new technology and infrastructure, upskilling of the labor force, and behavioral changes in the habits of both consumers, businesses and regulators. Simultaneously, coordination and collaboration between the public and the private sector must be enhanced.

GOVERNMENT'S CENTRAL ROLE

This is where the Uzbek government can and must play a central role. The transition to more a circular economy will not only help Uzbek withstand supply chain shocks but will also help attract investment in an era where ESG criteria continues to determine global capital allocation.

Institutional investors and sovereign wealth funds increasingly view high-waste, high-emission business models as long-term risks. Consequently, governments that proactively establish robust circular frameworks that facilitate private sector investment can see a higher degree of Foreign Direct Investment (FDI).

On the following pages this report will address water, waste, and the energy transition from a circular perspective, addressing why it is important, how it can be solved, and the current status for these topics in Uzbekistan.

THREE INTERDEPENDENT SYSTEMS

Water

Efficient water use reduces energy consumption across agriculture and industry

Waste

Waste-to-energy reduces landfill pressure and diversifies the power mix

Energy

Cleaner energy reduces the carbon intensity of water treatment



01

Water

From extraction and discharge to regeneration and reuse – closing the water loop.

WATER

Closing the *water loop*

Within the broader circular economy, water management means the transition from a traditional linear model of extraction, use, and discharge toward a regenerative cycle that preserves both the quality and availability of water. In a linear system, water is often treated as an infinite resource extracted from natural resources and returned as degraded effluent, not to be used again. Conversely, a circular water strategy focuses on minimizing freshwater withdrawals by “closing the loop” through advanced treatment and reuse. This involves implementing cascading systems where the output of one process serves as the input for another, ensuring that water quality is precisely matched to its specific industrial, human or agricultural purpose.

Non-potable water can in many instances be good enough for other processes such as manufacturing or agriculture use. By integrating decentralized treatment technologies and smart monitoring, this framework prevents “leakages” in the form of physical loss or quality degradation, seeking to detach industrial productivity from water scarcity.

For both emerging and developed economies transitioning to circular water systems is a critical component of long-term security for both people and businesses. As climate variability continues to impact traditional water supplies, countries that can optimize their water cycles reduce the risk of production shutdowns and stabilize cost for consumers. This shift requires investments in new infrastructure and behavioral changes in the habits of businesses and consumers, guided by strong government regulation.

CASCADING WATER SYSTEM – CIRCULAR APPROACH



As climate variability continues to impact traditional water supplies, countries that can optimize their water cycles reduce the risk of production shutdowns and stabilize cost for consumers. This shift requires investments in new infrastructure and behavioral changes guided by strong government regulation.

WATER

Uzbekistan: water under *pressure*

\$5bn

Estimated annual economic losses
from water scarcity

90%+

Of total water consumption
by the agriculture sector

Oct 2025

New Water Code enacted – water
designated national asset



CURRENT STATUS IN UZBEKISTAN

Uzbekistan is grappling with severe water scarcity, exacerbated by climate change and inefficient water management, particularly in its agriculture sector which accounts for over 90% of water consumption. This situation results in substantial economic losses, declining river flows from the Amu Darya and Syr Darya, and the environmental catastrophe of the Aral Sea. Projections indicate a significant increase in water deficit by 2030 and 2050, with current economic losses due to scarcity estimated at around \$5 billion annually.

Uzbekistan is advancing initiatives to improve water efficiency and promote reuse. Efforts include the widespread adoption of water-saving technologies like drip irrigation, the digitalisation of water management systems, and the exploration of treated wastewater for agricultural purposes. A new Water Code (October 2025) is a key regulatory development. It designates water as a national asset, establishes frameworks for rational use, protection, water abstraction quotas, and introduces penalties for violations. This code integrates principles of integrated water resource management and climate change adaptation, aiming for a more sustainable and resilient water future.

WATER CODE – OCTOBER 2025

- Designates water as a national asset
- Frameworks for rational use and protection
- Water abstraction quotas introduced
- Penalties for violations
- Integrates climate change adaptation principles

KEY INITIATIVES

- Widespread adoption of drip irrigation technologies
- Digitalisation of water management systems
- Treated wastewater exploration for agriculture
- Integrated water resource management principles
- Significant water deficit projected by 2030–2050

CASE STUDY • SINGAPORE

“Four National Taps” – Water *Self-Sufficiency*

Singapore’s “Four National Taps” strategy is a comprehensive and integrated approach to water resource management, designed to ensure a sustainable and resilient water supply for the island nation. The primary objective is to achieve water self-sufficiency and reduce reliance on external sources, particularly imported water, in the face of limited natural resources and growing demand. This strategy was first envisioned in the 1972 Water Master Plan and formally termed in 2004.

THE FOUR TAPS

01- LOCAL CATCHMENT WATER

Collecting and treating rainwater from an extensive network of reservoirs and drainage systems covering two-thirds of Singapore’s land area

02- IMPORTED WATER

Primarily from Malaysia, which has historically been a significant source but is being strategically reduced

03- NEWATER

High-grade reclaimed water produced by advanced treatment of used water, meeting up to 40% of non-domestic/ industrial demand and projected to meet 50% by 2060

04- DESALINATED WATER

Converting seawater into potable water, with multiple plants contributing to the supply and expected to meet 30% of demand by 2060

SUPPLY MIX TRAJECTORY (2060 TARGETS)

LOCAL CATCHMENT 30%

IMPORTED WATER (REDUCING) 20%

NEWATER – 2060 TARGET 50%

DESALINATED – 2060 TARGET 30%



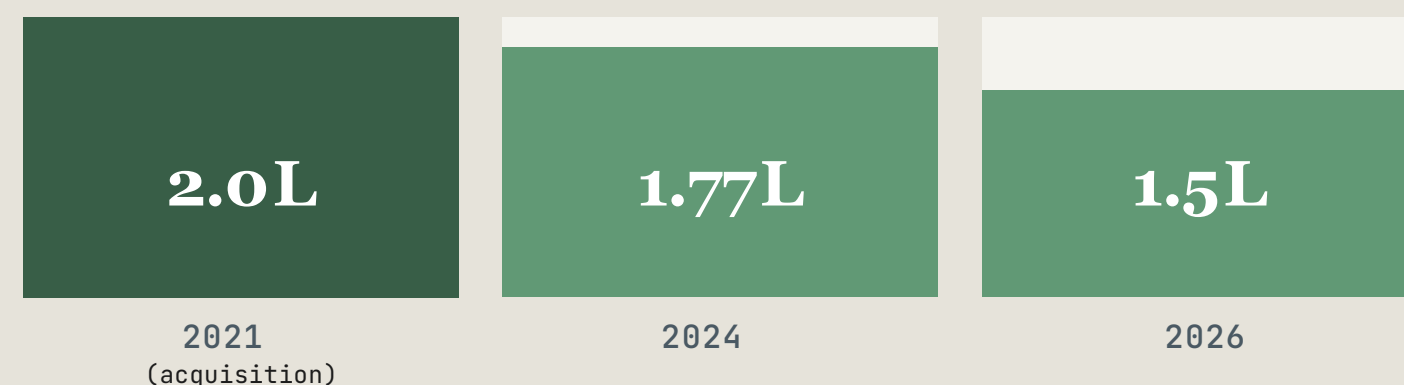
The overall significance of this strategy lies in its ability to transform Singapore from a water-vulnerable nation to a global leader in sustainable water management, demonstrating a closed-loop approach to water resources and providing a model for other water-stressed regions.

Water *Reuse* Systems by CCI

CCI Uzbekistan, the Coca-Cola Company's bottling partner in the country, has set voluntary internal targets to reduce the amount of water used to produce one liter of finished product. These targets, which go beyond national regulatory requirements, reflect the company's commitment to sustainable water stewardship as a core element of its circular economy strategy.

At the time CCI acquired and privatized the plant from the Government of Uzbekistan (CCBU) at the end of 2021, water consumption stood at 2 liters per 1 liter of finished product.

LITERS OF WATER PER LITER OF FINISHED PRODUCT



Through the progressive integration of sustainable water stewardship solutions, the company has reduced this figure to 1.77 liters in 2024 and 1.69 liters in 2025 with a target of 1.50 liters at the end of 2026.

CCI Uzbekistan strives to reuse and recycle every possible drop of water within its beverage production processes. Implementation of water reuse systems began at the Tashkent plant and has since been extended across the company's full production network. Starting from 2024, CCI Uzbekistan has started to use water recovery-reuse technologies in the facilities in order to reduce water consumption.

As a result of these combined efforts, CCI Uzbekistan is expected to save approximately 131 million liters of water in 2026.

~2.6 M

PEOPLE'S DAILY
WATER USE

Assuming
~50 litres/day
basic household use

52

OLYMPIC-SIZE
SWIMMING POOLS

1 pool ≈
2.5 million litres

~520 M

STANDARD
BOTTLES OF
WATERSAVED

Based on 250 ml bottles

~7,000

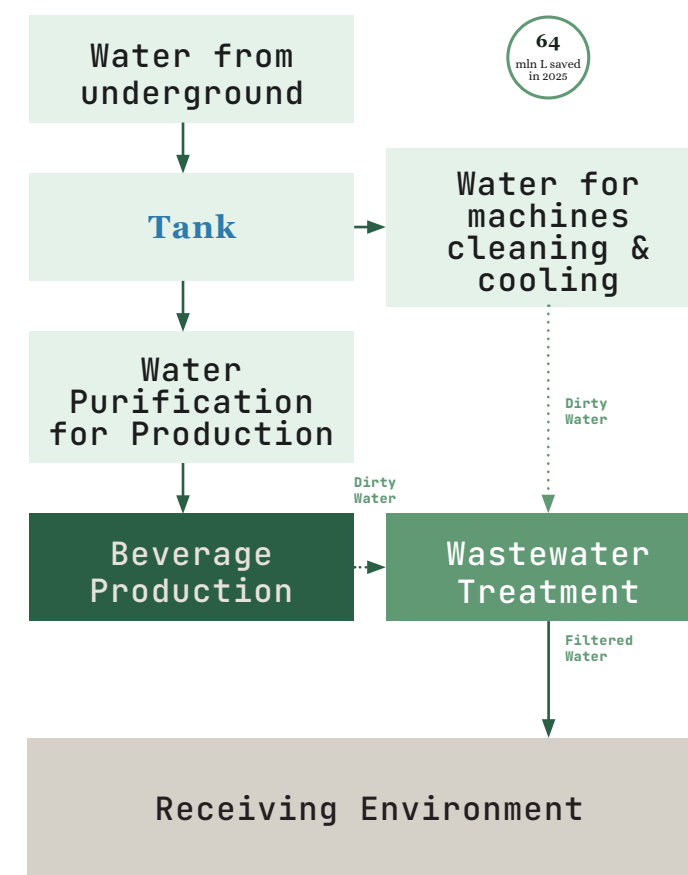
HOUSEHOLDS
FOR A FULL YEAR

Assuming
~50,000 litres/year
per household
for basic needs

WATER SYSTEM ARCHITECTURE

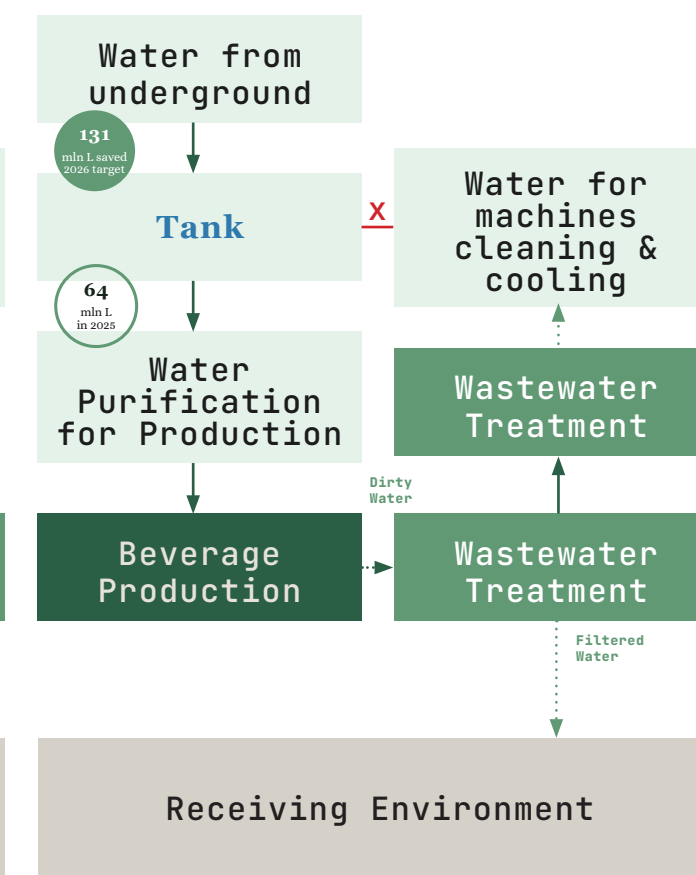
Before *vs.* After – How the loop was *closed*

BEFORE Water (Existing) – Linear Flow



Linear flow – water used once, wastewater discharged to environment

AFTER Water (New) – Circular Loop



Circular loop – wastewater treated and reused for machine cleaning, reducing fresh water intake by 25%



02

Waste

From linear disposal to a circular economy of materials — eliminating waste by design.



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NOWHERE IS LINEAR ECONOMY MORE EVIDENT THAN IN WASTE.

From waste to *resource*

Nowhere is linear economy more evident than in waste. A circular economy transitions waste from being an end-state pollutant to a potentially valuable resource stream temporarily misplaced. Moving away from the traditional “extract-use-dump” model requires a systemic overhaul of how materials are collected, classified, recovered, and eventually reintegrated into production and manufacturing. In this framework, products are ideally designed from the outset to be easily disassembled, repaired, or repurposed, ensuring that the energy and material embedded at first production are never lost.

By shifting the focus from simple disposal to high-quality recovery, the economy moves toward a state where the concepts of “waste” is all but eliminated, replaced by a continuous flow of technical and biological nutrients. This approach seeks to ensure that material remains within the economic loop for as long as feasible possible, radically reducing the volume of materials destined for landfills and decreasing the environmental footprint from primary extraction.

For emerging economies, modernizing waste systems is a critical lever. Not only for improving sanitation and human health but also for reducing vulnerability to global supply chain shocks but also for industrial resilience and the preservation of natural resources. A sophisticated resource-recovery sector stabilizes the domestic supply of raw materials, protecting manufacturers from the price volatility and supply chain disruptions of global commodity markets. Such a shift requires investment across the entire waste value-chain in additional to changes in consumer attitudes towards second-life products.

WASTE VALUE CHAIN — CIRCULAR APPROACH



WASTE

Uzbekistan: significant challenges, *decisive action*

7-14.8M

Tons solid household waste annually

<10%

Estimated recycling rate — majority to landfill

>10%

Of municipal waste is plastic

2025

Year of Environmental Protection & Green Economy

Uzbekistan faces significant challenges in waste management, with low recycling rates and a heavy reliance on landfills, generating around 7–14.8 million tons of solid household waste annually. Only a small percentage is estimated to be recycled. The majority ends up in landfills, many of which do not meet sanitation standards, leading to environmental degradation, soil and water contamination, and negative health impacts. Plastic waste is a growing concern, making up over 10% of municipal solid waste, with insufficient recycling infrastructure and low public awareness exacerbating the problem.

In response, Uzbekistan is actively pursuing strategies to tackle the issues. Key initiatives include the declaration of 2025 as the “Year of Environmental Protection and Green Economy” and the development of legal acts to improve waste management and promote the “zero waste” principle. Regulations such as a presidential decree mandating separate collection and disposal of household waste types are in place. The government is also introducing an Extended Producer Responsibility (EPR) scheme for plastic waste by 2026, aiming to hold manufacturers and importers accountable for recycling. Furthermore, an environmental fee on products like batteries, cars, and packaging is planned to boost recycling rates. New measures also include tax incentives and customs duty exemptions for enterprises involved in waste collection, sorting, and processing, and the establishment of an Agency for Waste Management and Circular Economy Development.

GOVERNMENT ACTIONS & INITIATIVES

YEAR OF ENVIRONMENTAL PROTECTION 2025

Declaration signaling national priority for green economy transition.

EPR SCHEME FOR PLASTICS (BY 2026)

Extended Producer Responsibility — manufacturers and importers accountable for recycling.

ENVIRONMENTAL FEE SYSTEM

Fee on products like batteries, cars, and packaging to boost recycling rates.

TAX INCENTIVES FOR WASTE SECTOR

Tax incentives and customs duty exemptions for enterprises involved in waste collection, sorting, and processing.

AGENCY FOR WASTE MANAGEMENT

Establishment of a dedicated Agency for Waste Management and Circular Economy Development.

ZERO WASTE LEGISLATION

Legal acts under development to improve waste management and promote the “zero waste” principle.

CASE STUDY • SWEDEN

Turning Trash into *Treasure*

Sweden’s waste management system is a global leader in circular economy practices, primarily driven by its “waste-to-energy” model and high recycling rates. The key objective is to minimise landfill waste, maximise resource recovery, and contribute to energy production. This is achieved through a comprehensive strategy that prioritises waste prevention, reuse, material recycling, and energy recovery, with landfilling as a last resort. Main components include extensive household waste sorting into multiple categories (e.g., paper, plastic, glass, organic waste), a robust network of recycling stations, and advanced waste-to-energy (WtE) incineration plants.

These WtE facilities convert non-recyclable waste into electricity and heat, powering district heating networks for millions of homes and businesses. Sweden’s success is underpinned by strong government policies, such as landfill bans on organic and combustible waste, extended producer responsibility (EPR) laws, and a landfill tax. Public participation is crucial, with high awareness and dedication among citizens leading to meticulous waste separation. As a result, less than 1% of Sweden’s household waste ends up in landfills, with approximately 47% recycled and 52% converted into energy. This integrated approach has transformed waste from a burden into a valuable resource, significantly reducing carbon emissions and providing a sustainable energy source.

<1%

To landfill

~47%

Recycled

~52%

WtE energy

KEY POLICY LEVERS

01-Prevention & Reduction

02-Reuse

03-Material Recycling (~47%)

04-Energy Recovery WtE (~52%)

05-Landfill—last resort (<1%)

KEY POLICY LEVERS

- Landfill ban on organic & combustible waste
- Extended Producer Responsibility (EPR) laws
- Landfill tax to incentivize alternatives
- Mandatory household waste sorting categories
- WtE plants for non-recyclable residuals
- Strong public awareness campaigns





03

Energy Transition

From fossil fuel dependency to a circular, renewable energy architecture.

FROM FOSSIL FUEL DEPENDENCY TO A CIRCULAR ENERGY ARCHITECTURE

A circular energy *architecture*

The transition towards a circular energy system centres on replacing finite, extractive fuel sources with a perpetual flow of renewable clean power. Unlike the linear energy model, which relies on the continuous depletion of fossil reserves, a circular framework treats the sun, wind, and water as the primary inputs for an infinite energy loop. This transition requires more than just installing solar panels; it involves a systemic redesign of the many fossil-centric grids to incorporate renewable sources which follow a different production pattern. In this model, renewable energy becomes the “circulating currency” that powers industrial activity. While the abatement of CO₂ emissions is a critical and necessary byproduct of this shift, the primary objective is to create an energy architecture that is self-sustaining and independent of the volatile cycles of resource extraction.

In this context, a robust renewable sector is the cornerstone of strategic autonomy and long-term industrial competitiveness. By domesticating energy production through wind, solar and geothermal assets, countries can shield their industries from the price shocks of global fuel markets and reduce the “energy leakage” associated with inefficient, centralized power plants. This shift requires large-scale investments in physical infrastructure, upgraded grid-capacity and the deployment of advanced demand-management technology which can handle the intermittent flow of clean energy sources.

Solar & Wind

Primary infinite inputs
replacing fossil fuel extraction
with perpetual natural cycles

Grid Modernisation

Redesigned networks to handle
distributed, intermittent
renewable sources

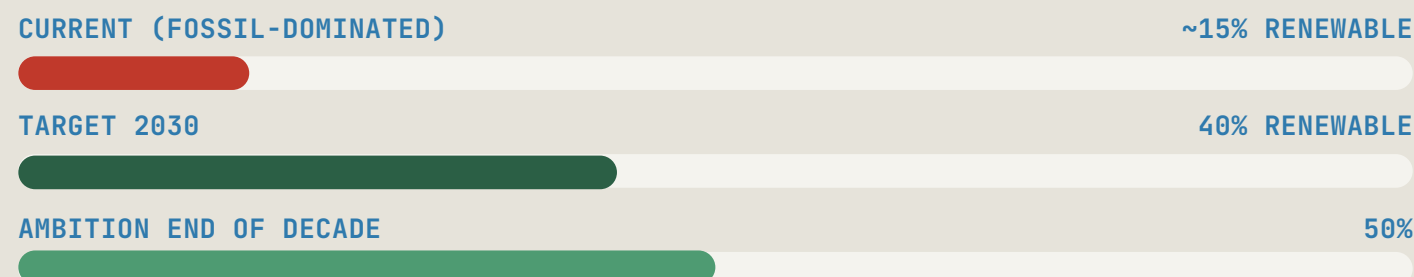
Energy Efficiency

Demand management technology
reducing total consumption from
primary resources

FROM FOSSIL FUEL DEPENDENCY TO A CIRCULAR ENERGY ARCHITECTURE

Uzbekistan: significant challenges, *decisive action*

RENEWABLE ENERGY ROADMAP



CURRENT STATUS IN UZBEKISTAN

Uzbekistan's energy sector is currently heavily reliant on fossil fuels, primarily natural gas and coal, leading to significant greenhouse gas emissions and resource depletion. The nation is actively pursuing an energy transition, with ambitious targets to increase the share of renewable energy in its generation capacity. Recent updates indicate a goal of 40% renewable energy by 2030, and potentially surpassing 50% by the end of the decade. Compared to the gap between the current renewable energy levels, these targets require a significant deployment effort in the coming years.

From a circular economy perspective, Uzbekistan is integrating waste management with energy production. A key initiative is the development of Waste-to-Energy (WtE) projects across multiple regions, aiming to convert municipal solid waste into electricity. Furthermore, Uzbekistan is focused on enhancing energy efficiency across all sectors, with a national strategy to halve energy use per unit of GDP by 2030, reducing overall demand for primary resources. Regulatory reforms and presidential decrees are attracting foreign investment in renewables and supporting these circular approaches.

SAMARKAND WTE FACILITY

1,500t **240M**
solid waste/day kWh electricity/year

A high-tech WtE facility in Samarkand, operational between 2025 and 2027, designed to process up to 1,500 tons of solid waste daily, generating approximately 240 million kilowatt-hours of electricity annually.

ENERGY EFFICIENCY TARGET

-50%
energy use per unit of GDP by 2030

A national strategy to halve energy use per unit of GDP by 2030, reducing overall demand for primary resources. Regulatory reforms and presidential decrees are attracting foreign investment in renewables.

CASE STUDY • KAZAKHSTAN

The *Carbon Neutral* Train of Central Asia

Kazakhstan has committed to achieving carbon neutrality by 2060, a significant transition for a country heavily reliant on fossil fuels, particularly coal, which currently accounts for around 70% of its energy mix. This ambitious goal is enshrined in its Strategy for Achieving Carbon Neutrality by 2060.

A central pillar of this transition is the aggressive expansion of renewable energy. Kazakhstan aims to increase the share of renewables in its power mix to 15% by 2030 and a substantial 50% by 2050. To achieve this, the country plans to add 10 GW of new renewable energy capacity by 2035, requiring an estimated \$20 billion in investment. Initiatives include modernising energy legislation, introducing transparent auctions to attract investment, and deploying energy storage solutions like pumped hydro and battery systems to integrate intermittent renewables. Notable projects include the 1 GW Mirny wind farm, which will feature a 600 MWh battery energy storage system. Kazakhstan is also exploring its significant potential in solar, wind, and nuclear energy, while gradually phasing out new coal-fired generation projects and converting existing ones to natural gas as a transitional fuel.

KAZ ENERGY TARGETS TIMELINE

2020 3% renewable target – achieved

2023 5.92% renewables actual electricity production

2030 15% renewables target

2035 26 GW new capacity; \$20bn investment required

2050 50% renewable and alternative sources

2060 Carbon neutrality target

KEY INITIATIVES

- Modernising energy legislation and transparent auctions to attract investment
- Deploying energy storage: pumped hydro and battery systems
- 1 GW Mirny wind farm with 600 MWh battery storage system
- Deals with ACWA Power, Masdar, TotalEnergies for 3 GW wind capacity
- Exploring solar, wind, and nuclear energy potential
- Gradual phase-out of new coal, converting to gas as transitional fuel
- National referendum 2024 voted in favour of nuclear construction



04

Country Comparisons

Learning from global leaders
and peer economies.

EU FRONTRUNNER

 Belgium

~11.8M

Population 2025

~€635bn

GDP

CE Leadership
Highest recycling rate in EU (54.7%);
Second highest CMUR (22.7%)

As part of the European Union (EU), Belgium is widely assessed as one of the frontrunners in the transition towards a circular economy, particularly within responsible waste and water management. Its relatively complex, decentralized, multi-level governance structure allows the various Belgian regions to operate with their own regional strategies. While separate strategies can promote autonomy and efficiency, the lack of a national circular economy strategy can also create large regional differences or inertia, thus to some extent counteracting potential benefits on a national scale.

Belgium's transition towards increased use of renewable energy is also negatively affected by the complex governance structure and regions operating independently. Compared to waste and water management, where investments are comparatively small and where the regions themselves reap most of the benefits, transitioning away from fossil fuel reliance and upgrading grid infrastructure requires substantial investments and collaboration on a national or cross-regional scale.

54.7%

Recycling rate EU (2022) – #1

22.7%

Circular Material Use Rate
(CMUR) 2024

700 kg

Waste per capita
(4th highest in EU/EEA)

Belgium – *Water & Waste*

WATER

Belgium faces high levels of baseline water stress and is often ranked among the top 25 countries globally when considering freshwater withdrawal as a proportion of available freshwater resources. While causes differ between regions, climate change and an aging infrastructure are common denominators for Wallonia, Flanders, and Brussels. For Brussels and Flanders, high population density and widespread concrete paving are also contributing factors.

This imbalance between freshwater withdrawal and availability continues to shape how Belgians and Belgian regions brace themselves for a possible future where freshwater is a scarce resource. At the same time, Belgium's water-use efficiency and its high proportion of safely treated wastewater are signs that Belgium is working to mitigate some of the consequences originating from the high-water stress levels, focusing on national and regional policies and initiatives.

The summer of 2021 was a turning point for the awareness around responsible water usage, both on a national and individual level. Excessive floods and severe droughts provided clear evidence that Belgium's current water management had to be adjusted and adapted to a future where effects from climate change will only exacerbate water stress levels.

Going forward, Belgium must continue its collaboration between regions on a policy and action level. Responsible water management with particular emphasis on freshwater is key to overcome the challenges climate change and an aging infrastructure pose.

WASTE

Belgium is a circular economy leader in waste management and material recovery, boasting one of the highest recycling rates in the EU (54.7% in 2022) and strong secondary material markets with a CMUR of 22.7% in 2024. Up to 95% of metals are recycled, with recycling rates for paper, cardboard, and glass between 80% and 85%.

These recycling rates are combatting the relatively large volumes of waste generation in Belgium. With approximately 700 kg waste generated per capita (2024), the fourth highest in the EU/EEA area behind Austria, Denmark and Norway, recycling and reintroducing materials in the economy is essential to reduce the environmental footprint.

Among the actions and initiatives to reduce waste generation, the Flanders Local Materials Plan 2023–2030 target of a 30% reduction of reduce household residual waste by 2030 remains ambitious. Wallonia has launched its ambitious Circular Wallonia plan, containing more than 60 measures to transition from a linear economy to a zero-waste system. This policy also addresses the job market, with a target of increasing circular economy jobs by 20% by 2025.

Working strategically with certain sectors remains a priority for the Belgian regions. Several initiatives targeting circularity in the construction sector, electronics, and food are ongoing, including subsidies for circular construction projects, the Flanders Recycling Hub, and Wallonia's REGAL Plan.

95%

Metals recycled

80–85%

Paper & glass
recycled

60+

Wallonia CE
measures recycled

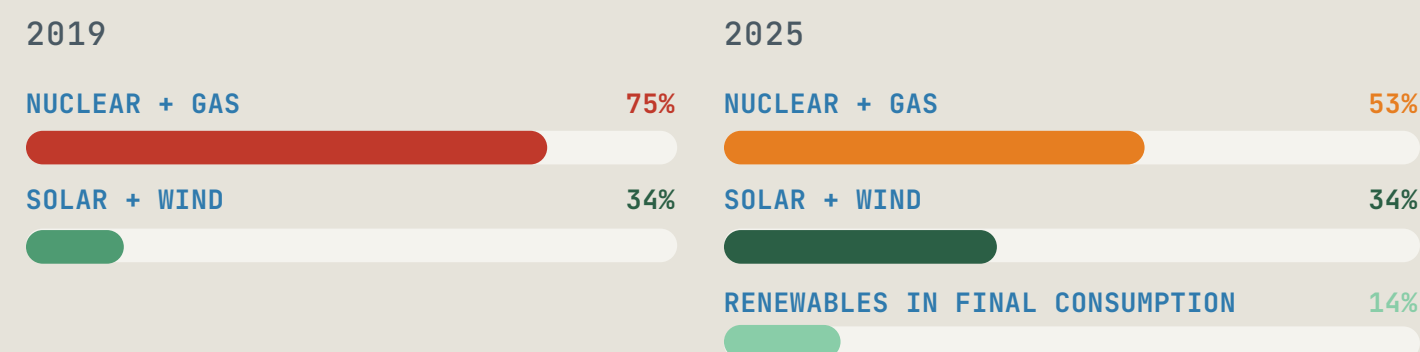
Belgium – *Energy Transition*

Following the momentum of the European Union, Belgium is now continuing to add renewable energy sources in their electricity mix. Where nuclear power and gas traditionally have been the main energy sources, comprising 75% of the electricity generation mix as late as in 2019, the same sources now account for 53% (2025), with the majority originating from nuclear power. In comparison, solar and wind power generation has increased from 13% to 34% in the same period. Combined with the strong regulatory push from EU, the energy crisis in 2022 has contributed to reduce Belgium's gas consumption and increased their energy resilience.

Given Belgium's frequent status as a net electricity importer, the energy mix is also affected by electricity generation in neighboring countries, in particular France and the Netherlands. Where the former has good availability of nuclear power, the latter contributes with wind power at times of abundant wind generation in Northern Europe and Germany. Further integration with the European grid will be important going forward for Belgium's energy transition.

Positive development in electricity generation aside, Belgium's final energy consumption coming from renewable sources remains an EU low, with approx. 14% (2024). Geographic factors such as high population density and space constraints and a relatively short coast limiting offshore wind, combined with an energy-intensive manufacturing sector and households locked into natural gas and oil heating are main contributors.

BELGIUM ELECTRICITY MIX – SHIFT 2019 → 2025



Belgium demonstrates both the potential and the limits of the EU renewable transition in a dense industrial economy. The governance fragmentation between regions is the primary structural impediment to accelerating its energy and circular economy transition.

OECD INDUSTRIAL ECONOMY

United Kingdom

~69.5M

Population 2025

~£4,000bn

GDP

CE Leadership
First country with legally binding
net zero target (2050); leader in
offshore wind capacity

The United Kingdom is widely recognised as one of the world's frontrunners in the transition towards a low-carbon and increasingly circular economy. Its legislative framework is considered pioneering, in particular the Climate Change Act of 2008, which established the world's first legally binding long-term climate target. While the UK's energy transition has delivered impressive and measurable results, structural challenges persist in water management and waste, where practical outcomes have not always kept pace with political ambition.

The UK's departure from the European Union in 2020 has introduced a degree of regulatory divergence, as the country is no longer bound by EU directives. However, the majority of EU environmental standards were retained in domestic law through the European Union (Withdrawal) Act 2018, and the subsequent Environment Act 2021 established a domestic equivalent of EU-level environmental governance. The challenge going forward is whether the UK can maintain regulatory momentum outside the EU framework, and whether England, Scotland, Wales, and Northern Ireland together can coordinate effectively in the absence of a binding national circular economy strategy.

127 TWh

Record renewable electricity
2025 (~44% of supply)

33 GW

Total wind capacity Q3 2025

44.6%

Household recycling rate
2023 (below EU 50% target)

United Kingdom – *Water & Waste*

WATER

The United Kingdom faces a growing and increasingly visible water management crisis, rooted in infrastructure that is among the oldest in Europe. With almost 20% of water lost by water companies before reaching household taps, large investments are needed to preserve valuable water reserves. The old infrastructure also impacts England's combined sewer systems. These were originally designed to collect both domestic wastewater and surface runoff. Under conditions of heavy rainfall, the systems are deliberately designed to overflow, releasing untreated sewage directly into rivers and coastal waters. This mechanism and its effects have in recent years become a focal point of public and regulatory concern. In 2024, English water companies discharged untreated sewage for a record 3.614 million hours into rivers and coastal waters, a figure that had more than doubled compared to 2022 levels of 1.75 million hours. Research has demonstrated that chronic underinvestment in England's sewage treatment infrastructure, combined with population growth and the effects of climate change on rainfall intensity, is the primary driver.

Against this backdrop, the UK Government introduced the Water (Special Measures) Act, which received Royal Assent on 24 February 2025. The legislation significantly strengthens the enforcement powers of the Environment Agency and Ofwat, enabling stricter penalties, criminal prosecutions against company executives, and requiring water companies to publish annual pollution reduction plans.

Looking ahead, the challenge for the UK is substantial. South-East England is already under serious water stress, and several regions are projected to face severe water scarcity by 2030.

3.614M hours

Untreated sewage discharged 2024 — record high

WASTE

The United Kingdom has made meaningful progress in reducing its dependence on landfill disposal over the past two decades, driven by escalating Landfill Tax rates and EU-derived targets that were subsequently embedded in domestic law. The volume of biodegradable municipal waste sent to landfill fell from 6.3 million tonnes in 2022 to 5.3 million tonnes in 2023. In the construction and demolition sector, the UK performs well, with 59.4 million tonnes of non-hazardous waste recovered from a total of 63.0 million tonnes generated in England in 2022.

The overall household recycling rate, however, tells a more mixed story. UK's provisional household recycling rate was 44.6% in 2023, which is below the EU's binding minimum target of 50% and far behind leading EU member states such as Germany. Wales is a notable outlier, having reached a recycling rate of 57.0% in 2023, driven by over a decade of mandatory food waste collections.

The absence of a national, harmonised circular economy strategy for all four UK nations is a structural weakness. Policy approaches diverge significantly between England, Scotland, Wales, and Northern Ireland, creating inconsistency in collection systems, infrastructure investment, and public guidance. The long-delayed deposit return scheme for beverage containers is illustrative of this challenge.

A further concern is the reliance on energy-from-waste (EfW) incineration as the primary alternative to landfill disposal. While EfW diverts waste from landfill, it does not constitute recycling, and expanding EfW capacity could create a structural disincentive to invest in genuine circular economy outcomes.

United Kingdom – Energy *Transition*

The United Kingdom's energy transition is, by international comparison, one of the most advanced among major industrialised economies. The country has moved with remarkable pace from a system heavily reliant on coal and gas to one in which renewable energy sources, mainly wind, both onshore and offshore, supply approaching half of national electricity generation. In 2025, renewable sources produced a record 127 terawatt-hours (TWh) of electricity, representing approx. 300% of the 43 TWh generated by renewables in 2015, and accounting for approximately 44% of total electricity supply.

The UK's offshore wind sector is a particular source of strength and international leadership. Total wind capacity reached 33 GW in the third quarter of 2025, and one of the world's largest wind farms, Dogger Bank A, has been delivering power since autumn 2025, with Dogger Bank B due to follow in 2026. The UK's Government target of 50 GW of offshore wind capacity by 2030 underpins an ambitious industrial and clean energy strategy. September 2024 also marked a historic milestone with the closure of Ratcliffe-on-Soar, the UK's last remaining coal-fired power station, making the UK the first G7 nation to fully eliminate commercially operational coal-fired power plants.

The legislative foundation for the UK's energy ambition was laid by the Climate Change Act 2008, amended in 2019 to set a legally binding net zero target for 2050, making the UK the first G7 nation to do so. The Energy Act 2013 introduced the Contracts for Difference (CfD) auction mechanism, which has provided long-term price certainty to renewable energy investors and driven substantial cost reductions across wind and solar technology.

There are still several significant challenges to the energy transition. The carbon intensity of UK electricity marginally increased in 2025, reaching 126 gCO₂/kWh, partly because the retirement of coal generation was offset by increased gas-fired power, and nuclear output fell to its lowest level in half a century at 36 TWh. All but one of the UK's existing nuclear power stations are scheduled for decommissioning by 2030. Grid infrastructure represents a further structural bottleneck.

127 TWh

Record renewables 2025
(~44% supply)

33 GW

Wind capacity Q3 2025

50 GW

Offshore wind target 2030

September 2024: Ratcliffe-on-Soar closed – UK's last coal-fired power station. First G7 nation to fully eliminate commercially operational coal power.

EMERGING MARKET ECONOMY

Romania

~19M

Population 2025

~\$428bn

GDP

CE Leadership
Launched world's largest national
DRS (3bn returns in year one, 5.2bn
by end 2025)

Romania is an EU member state navigating a significant structural transition from a heavily centralised, fossil fuel-reliant, and landfill-dependent economy towards the higher standards required by EU environmental law. Since accession in 2007, Romania has transposed the bulk of the EU's environmental directives and regulations into national law. The gap between legislative adoption and practical implementation, however, remains one of the largest in the Union.

In waste management, Romania holds the EU's lowest municipal recycling rate and faces active infringement proceedings from the European Commission. In water, the urban–rural divide in wastewater treatment is acute, and investment in infrastructure, while ongoing with EU support, has been chronically delayed. In energy, the picture is more optimistic, where renewables accounted for approx. 49% of electricity generation in 2023, driven primarily by hydropower, and a dramatic expansion in solar capacity is now under way.

The overriding challenge for Romania is one of implementation pace. Legislative frameworks exist, EU funding is available, and targets have been set. Administrative capacity, governance quality, and enforcement have, however, consistently fallen short of what is needed to close the gap between ambition and outcome.

74%

Landfill rate 2022 (EU
avg 24%)

49%

Electricity from
renewables 2023

5.2bn

DRS containers
returned by end 2025

84%

DRS collection rate
October 2024

Romania – *Water & Waste*

WATER

Romania's water sector is characterised by a fundamental urban–rural divide and a decades-long underinvestment in basic infrastructure that continues to place it among the lowest performers in the European Union on sanitation and wastewater treatment. The country possesses substantial freshwater resources, with water stress levels at just 7.6%. The challenges lie not in the quantity of water available but in the quality of the infrastructure used to collect, distribute, and treat it.

In 2022, only 31% of domestic wastewater in Romania was safely treated, with 38% of industrial wastewater receiving some form of treatment. This figure is far below the EU average and reflects a structural deficit in sewage collection and treatment capacity, particularly in rural areas. Of the total domestic wastewater flow, 59% originates from sewers, 1.6% from septic tanks, and 40% from other sanitation facilities.

Romania has received substantial EU funding to address this infrastructure gap through successive Cohesion Policy periods since 2007. However, the implementation record has been poor. Projects have been delayed by corruption, poor contractor performance, administrative shortcomings, and the financial vulnerability of local authorities. Romania has been subject to multiple EU infringement proceedings related to the Urban Wastewater Treatment Directive. Water-use efficiency stands at \$25/m³.

Going forward, Romania has firm plans to extend water and wastewater infrastructure through the current EU programming period, supported by the EBRD and other international financial institutions.

31%

Domestic wastewater safely treated (2022) – far below EU average

WASTE

Romania's waste management performance is the most acute structural challenge it faces on its path toward a circular economy. Romania has one of the weakest recycling rates since 2010 in the EU. The landfill rate was 74% in 2022, with no progress made since 2010. As late as in December 2023, the European Court of Justice ordered Romania to pay €1.5 million for failing to comply with obligations regarding 31 unauthorised landfills.

Romania took a bold step by launching a nationwide Deposit Return System (DRS), branded as RetuRO SGR, on 30 November 2023, covering single-use plastic bottles, glass bottles, and aluminium cans for beverages. In just one year of operation, more than 3 billion containers had been returned, with the national collection rate reaching 84% in October 2024. By the end of 2025, Romanians had returned more than 5.2 billion containers under the deposit-return system. This represents a remarkable consumer behaviour shift in a very short period.

The DRS is a targeted, high-performance mechanism for beverage containers specifically, demonstrating that well-designed economic incentives can achieve swift changes in collection behaviour. In this regard, it is essential to recognise that EPR and DRS are not mutually exclusive or sequential, but rather complementary instruments.

It is essential with a smooth and well-managed transition from a purely EPR-based system to one in which DRS handles the high-volume beverage container stream, while EPR retains responsibility for the remaining packaging categories.

5.2bn

Containers returned end 2025

84%

Collection rate
Oct 2024

Romania – *Energy Transition*

Romania's energy transition is at an earlier and more complex stage than most Western European peers. The trajectory, particularly in renewable electricity, is increasingly positive. The country benefits from a diversified natural resource base, with meaningful potential in hydropower, wind, solar, and to a lesser extent biomass. In 2023, renewable sources produced 49% of Romania's electricity: hydropower contributed 31%, wind 13%, and solar 4%, with the remaining 51% supplied by fossil fuels and nuclear power.

Romania has set a target of achieving a 30.7% share of energy from renewable sources in gross final energy consumption by 2030, compared with 24.4% achieved in 2020, with the latest available data showing an increase to 25.8% in 2023. However, the EEA has assessed that this target will not be reached following the historical trend, and that acceleration is needed.

The legislative and policy framework for Romania's energy transition has been shaped primarily by its EU membership obligations. The National Energy and Climate Plan (NECP) 2021–2030 sets the overarching strategic direction, and Romania has benefited from EU funding mechanisms including the Modernisation Fund and the Recovery and Resilience Facility. One of the major challenges remains the Oltenia Energy Complex (CEO), which is Romania's largest coal producer and power generator.

In 2023, GHG emissions per capita in Romania were 3.28 tCO₂e, below the EU average of 5 tCO₂e. In the same year, the GHG intensity of GDP was 495 gCO₂e/EUR, above the EU average of 225 gCO₂e/EUR. The elevated GHG intensity of GDP reflects Romania's still energy-intensive economic structure.

The key structural challenges that Romania must address in its energy transition include the managed exit from coal and the integration of a rapidly growing solar fleet into a grid designed for centralised hydro and thermal generation. Also integral is the roll-out of large-scale energy storage capacity and the reform of the electricity market.

ROMANIA ELECTRICITY MIX 2023



49%

Electricity from renewables 2023

3.28 t

GHG per capita 2023
(below EU avg 5t)

30.7%

2030 renewables target in
gross final consumption

PEER ECONOMY – CENTRAL ASIA

Kazakhstan

~20.5M

Population 2025

~\$291bn

GDP

CE Leadership
First in Central Asia to launch a
national ETS. Carbon neutrality
target by 2060

Kazakhstan is Central Asia's largest economy and, by many measures, one of its most strategically significant. It exhibits all the characteristics of carbon lock-in by being dependent on exports of oil and gas, while its abundant and inexpensive coal is the main fuel for the power generation sector, amounting to approx. 70%. Kazakhstan is among the world's largest producers of uranium, chromite, and a range of critical minerals, which could prove decisive in the global clean energy transition.

At the same time, Kazakhstan has demonstrated a genuine, if uneven, commitment to reform. It was the first country in Central Asia to launch a national emissions trading system, introduce a functioning support mechanism for renewables, and set a carbon neutrality target. The achievements signal a political direction of travel that deserves recognition even where implementation has fallen short.

Policy coherence and governance quality remains structural challenges for Kazakhstan. While legislation exists for extended producer responsibility, emissions trading, and for renewable energy auctions, its implementation is hampered by energy subsidies that undermine market incentives and a government-led EPR operator that lacks the transparency and efficiency characteristics of industry-led approaches.

The next decade will determine whether Kazakhstan can convert its ambitions and its natural resource endowment into a genuine structural transition.

70%

Coal share of
electricity generation

7.7%

Fossil fuel subsidies
as % of GDP

5.92%

Renewables actual
share of electricity
2023

26 GW

New energy capacity
target by 2035

Kazakhstan – *Water & Waste*

WATER

Kazakhstan's water situation is shaped by a stark geographic paradox. The country is the ninth-largest in the world by land area, but also largely arid, with long-term average annual precipitation hovering around 250–300 mm per year. Renewable water resources stand at 3,212 m³ per capita (2022), and water withdrawal reaches 1,310 m³ per capita. As a result, 35% of Kazakhstan's renewable water resources are being withdrawn, placing the country in the medium-low water stress category and well above the global average of 18%.

Wastewater treatment is a significant challenge. Only 30% of domestic wastewater in Kazakhstan is safely treated, and just 2% of industrial wastewater receives any form of treatment. 49% of monitored water bodies in Kazakhstan have good ambient water quality, indicating that water pollution from inadequately treated wastewater and industrial discharges is a genuine environmental risk. In contrast, 92% of the population uses a safely managed drinking water service.

The Aral Sea remains the defining symbol of Kazakhstan's water governance legacy. The near-total desiccation of the once fourth-largest lake in the world remains one of the worst environmental disasters of the twentieth century. Kazakhstan has made notable progress in partially restoring the North Aral Sea through the Kokaral Dam, completed in 2005 with World Bank support, which has raised water levels and improved local conditions.

Kazakhstan's water challenges are intensifying due to climate change. Rising temperatures are accelerating glacial retreat in the Tian Shan and Altai mountain ranges, which serve as the primary water towers for the country's major rivers. Investment in modern irrigation technology, water pricing reform, and wastewater treatment infrastructure represents the most critical near-term priority.

WASTE

Kazakhstan's waste management sector is characterised by a gap between official reporting and ground-level reality. Officially, 24% of the 4.1 million tons of municipal waste produced in 2023 was described as recycled – yet only 1.1% of all municipal waste results in the production of secondary raw materials. In 2024, approx. 37% of sorted municipal waste was still ultimately landfilled.

The physical scale of waste accumulation is striking. The country has already amassed 48.3 million tons of solid waste. Of approximately 3,000 landfills in operation, only 20.6% meet national standards, and 4,886 unauthorised landfills have been identified. In seven of Kazakhstan's 20 largest cities, landfill capacity is nearing exhaustion.

Kazakhstan's EPR model is administered through a single government-controlled national operator (JSC Zhasyl Damu). While centralised accountability ensures equal coverage of waste streams, the model lacks the eco-design incentives characteristic of industry-led schemes such as Germany's Duales System Deutschland or Belgium's Fost Plus. When EPR is administered by a government operator rather than an industry-led Producer Responsibility Organisation (PRO), revenue flows without transparent reporting, and recycling targets become reporting exercises rather than genuine performance benchmarks.

The Kazakh government has approved 94 investment projects to construct waste processing plants – a meaningful step. But processing capacity alone will not solve Kazakhstan's waste challenge without accompanying reform to the EPR governance structure, meaningful landfill taxes, and investment in collection infrastructure.

48.3M t

Solid waste accumulated

4,886

Unauthorized landfills identified

Kazakhstan – Energy *Transition*

Kazakhstan's energy transition is at one of the earlier stages among countries that have formally committed to carbon neutrality. Coal accounts for approximately 70% of electricity generation and is indispensable for residential heating and heavy industry. Fossil fuel subsidies amount to approx. 7.7% of GDP, weakening incentives for energy efficiency investment and making it structurally difficult for renewables to compete on cost. Still, Kazakhstan has built an increasingly credible renewable energy policy architecture. The 2013 Concept of the Transition to the Green Economy set targets of 3% renewables by 2020, 15% by 2030, and 50% by 2050. In 2018, renewable energy auctions were introduced, driving tariffs for wind and solar to under \$0.030 per kWh.

In 2023, renewables accounted for just 5.92% of total electricity production. By 2035, the country targets 26 GW of new energy capacity, predominantly wind and solar. Kazakhstan has signed deals with ACWA Power, Masdar, and TotalEnergies for 3 GW of wind capacity with integrated storage systems.

Grid integration remains the defining technical challenge. Kazakhstan's power system is split into three zones with limited interconnectivity, dominated by inflexible coal-fired plants with insufficient balancing capacity. The future of coal is Kazakhstan's most consequential energy policy decision. Aging Soviet-built thermal plants are experiencing increasingly frequent accidents and blackouts — yet in April 2024, Kazakhstan signed a cooperation agreement with Russia on the construction of three new thermal power plants. This risks locking Kazakhstan into carbon-intensive infrastructure for decades, directly contradicting its 2060 carbon neutrality pledge.

Finally, Kazakhstan is the world's largest uranium producer but has no nuclear capacity in operation. A national referendum held in autumn 2024 returned a majority in favour of nuclear power plant construction — which, if pursued, could provide the low-carbon baseload needed to make Kazakhstan's renewable integration path viable.

70%

Coal share electricity
generation

5.92%

Renewables actual share 2023

26 GW

New capacity target
by 2035

Kazakhstan is the world's largest uranium producer. A 2024 national referendum voted in favour of nuclear power construction — which could provide the low-carbon baseload needed to make Kazakhstan's renewable integration path viable.



ESG REVIEW REPORT

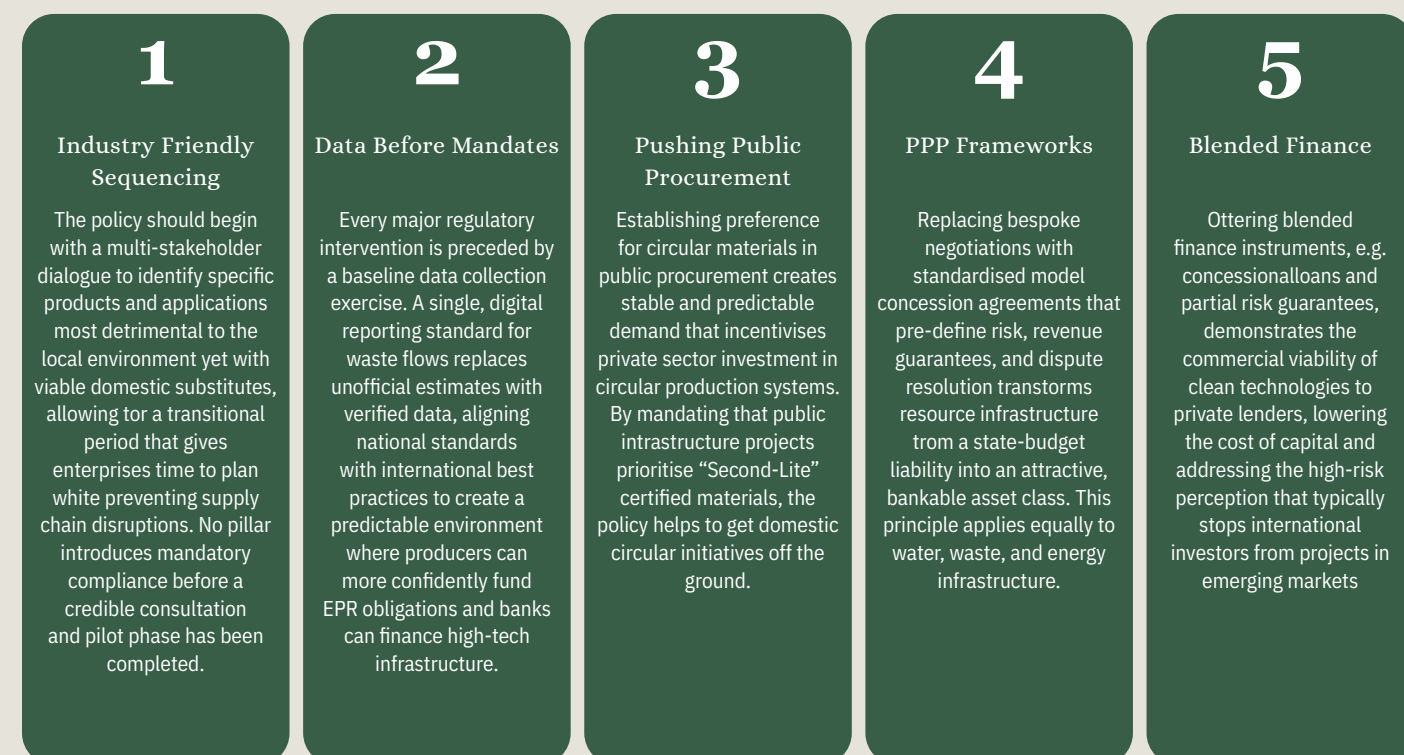
Roadmap and *Transition Design*

Uzbekistan – 2026-2046

Roadmap & *Transition* Design

Uzbekistan's CO2 emissions per unit of GDP are more than twice the OECD average, its water productivity remains among the lowest in the world, and municipal solid waste recycling has only recently risen to approximately 22% of solid waste, well below international standards. Yet these indicators of structural inefficiency are also indicators of structural opportunity. With annual economic growth projected at 6–7% and a reform-minded government, Uzbekistan is at an inflection point where the design choices made today will determine the resource intensity, and thus the long-term competitiveness, of an economy for decades to come.

This roadmap sets out a sequenced, phased, and industry-friendly pathway for Uzbekistan's transition toward a circular economy across three interdependent pillars: water, waste, and energy. Spanning 20 years from 2026 to 2046, it is organized into four phases: Foundation, Acceleration, Consolidation, and Maturity. Each of the phases build on the last. The approach prioritizes enablers before mandates, pilots before national rollout, and public-private partnerships as the primary vehicle for infrastructure delivery. The goal is not compliance for its own sake, but a structurally more competitive, resource-efficient, and investment-attractive economy. There are some cross-cutting principles which are relevant for all pillars and all phases. These are prerequisites for any positive development within circularity and the transition to a circular economy.

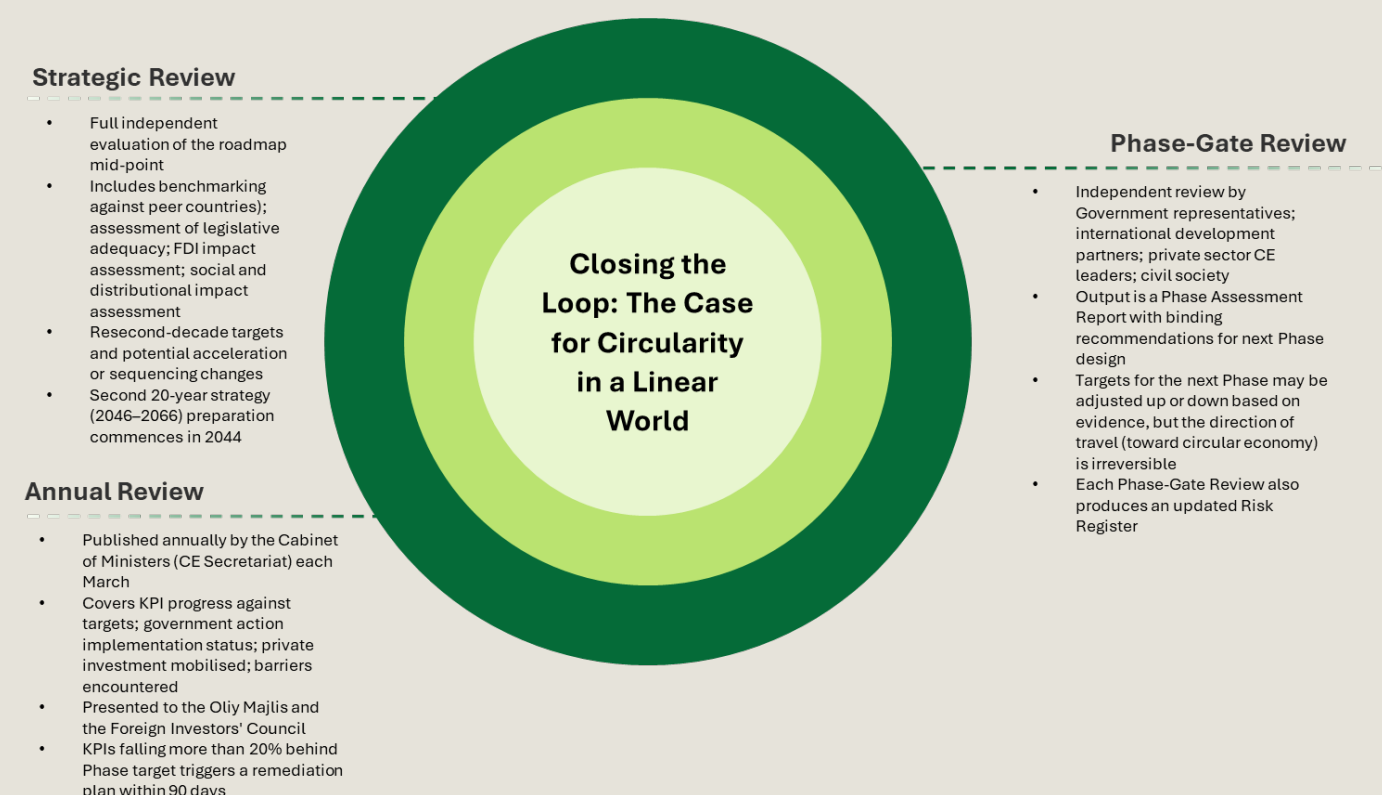


An comprehensive and detailed overview of proposed circular economy initiatives, identified problems related to the current linear economy, including detailed descriptions, current legislation, and anticipated outcomes of the proposed initiatives can be found in the Annex.

Governance

Establishing proper oversight and a robust review cycle is essential to maintain alignment between the roadmap and evolving economic, technological, and environmental conditions. Such a governance mechanism can be constructed in numerous ways. Regardless of structure and organisation, all circular economy pillars should have a simplified review annually, a phase-gate review at the end of each phase, and a ten-year strategic review halfway in the strategy period.

Below follows a proposed breakdown for the various review cycles, including format, participation, content, and reporting:



Objectives

Setting targets, measuring progress, and achieving goals are key components to ensure that the transition from a linear economy to a circular one becomes a success. For each of the four phases, the following KPIs and respective proposed targets can act as the blueprint for achieving the desired outcomes. These KPIs are also used in the country-by-country benchmarking to accommodate comparisons between peers and economies with a headstart on the circularity transition.

Water					
Indicator	Baseline (2025)	Phase 1 (2028)	Phase 2 (2032)	Phase 3 (2038)	Phase 4 (2046)
SDG 6.4.2 Water stress (%)	~122%	Establish measurement framework and publish water stress report	<110%	<70%	<50%
SDG 6.4.1 Water-Use Efficiency (USD/m ³)	~2.6 USD/m ³	ZLD pilots demonstrate 30% improvement in pilot facilities	~3.4 USD/m ³	~4.2 USD/m ³	>10 USD/m ³
SDG 6.3.1 Wastewater Safely Treated – Household (%)	~32%	Baseline survey completed and first PPP concessions awarded	~40%	~55%	>70%

Waste					
Indicator	Baseline (2025)	Phase 1 (2028)	Phase 2 (2032)	Phase 3 (2038)	Phase 4 (2046)
National Recycling Rate for Municipal Waste (%)	~5%	~10%	~20%	~32%	>40%
Circular Material Use Rate (%)	Not formally measured	Baseline methodology established and first CMUR calculated	~4%	~8%	~12%
Total Waste Generation (kg/capita)	~320 kg/capita	~320 kg/capita	~290 kg/capita	~250 kg/capita	~170 kg/capita

Energy transition					
Indicator	Baseline (2025)	Phase 1 (2028)	Phase 2 (2032)	Phase 3 (2038)	Phase 4 (2046)
SDG 7.2.1 Renewable Energy Share - Final Consumption (%)	~1%	~5%	~10%	~20%	>30%
SDG 7.3.1 Energy Intensity (MJ/USD)	~6 MJ/USD	~5.5 MJ/USD	~5 MJ/USD	~4 MJ/USD	<3 MJ/USD
GHG Emissions (tCO ₂ e/capita)	~3.5 tCO ₂ e/capita	Baseline MRV system established; first verified national GHG inventory	~3 tCO ₂ e/capita	~2.3 tCO ₂ e/capita	~1.5 tCO ₂ e/capita

Pillar 1: Water

KEY BARRIERS FOR IMPROVING WATER EFFICIENCY:

- Inadequate price signals in water markets create two critical inefficiencies: They fail to incentivise consumers and industrial users to reduce consumption during periods of low supply, and they distort investment signals for infrastructure upgrades and expansion.
- Water infrastructure investment in emerging economies often faces three primary barriers. One is related to regulatory and political risk originating from uncertain legal and taxation frameworks, while another is information asymmetry regarding local basin management practices, supply reliability, and demand forecasts. Ultimately, concerns about local autonomy over natural resources affects long-term project viability.
- High-quality drinking water is wasted for processes such as manufacturing or industrial cooling where lower-quality water would work perfectly well, potentially wasting premium water resources that could be consumed by humans.

PHASE 1: FOUNDATION

Among the relevant actions and initiatives for building the structural foundation for water efficiency in Uzbekistan are establishing credible water pricing signals, launching industrial zero liquid discharge (ZLD) pilots, and operationalising the October 2025 Water Code through implementing regulations.

Water – Phase 1: Foundation (2026 – 2028)		Proposed Actions and Objectives	
Government Actions	Instruments	Safeguards	Private-Public Cooperation
- Enact Water Code implementing regulations as soon as possible - Commission national water basin mapping with real-time monitoring infrastructure - Launch multi-stakeholder consultation on scarcity-indexed tariff design - Identify 3–5 pilot industrial facilities for voluntary zero-liquid discharge (ZLD) adoption	- Dynamic tariff framework that automatically adjusts as regional water stress fluctuates, with one tariff for households and another for high-volume consumers such as factories and industrial plants - World Bank / Asian Development Bank (ADB) concessional lending for basin monitoring infrastructure - Public-Private Partnership (PPP) concession template development	- Household tariff ring-fencing to prevent social harm - Public consultation requirement before any tariff adjustment - Independent tariff review board	- International technology firms (precision irrigation, digital water metering) invited into pilot procurement under existing PPP law - Standardised model concession agreements pre-defining risk, revenue guarantees, and dispute resolution to be drafted as template for water infrastructure
Responsible Entities and Enablers - Ministry of Water Resources - Cabinet of Ministers - National Committee for Ecology and Climate Change - International development partners (ADB, World Bank)			

Water – Phase 1: Foundation (2026 – 2028)

Risk Register

Risk	Potential Mitigation Measures
Political resistance to water tariff reform	- Phased introduction - Household tariff ring-fenced - Subsidies for low-income households

Water – Phase 1: Foundation (2026 – 2028)

Potential Barrier Removal Actions

Absence of Price Signals	Information Asymmetry	Regulatory/Political Risk for Investors	Industrial Inefficiency
- Legislate water abstraction quota system under the Water Code - Introduce basin-level water stress indices as published quarterly data	- Commission a national water data platform (open access) - Partner with International Water Management Institute (IWMI), GIZ or similar organizations for hydrological modelling	- Publish standardised PPP concession template for water - Commit to 10-year tariff stability guarantee in investment contracts	- Establish zero-liquid discharge (ZLD) pilot programme - Invite 3–5 factories in water-intensive industries (textiles, chemicals) to the ZLD pilot with a % capital grant co-financing

PHASE 2: ACCELERATION AND SCALE-UP

Among the relevant actions and initiatives for accelerating and scaling up the water as a circular resource in Uzbekistan are implementing scarcity-indexed tariffs nationally, expanding ZLD mandates to all new large industrial facilities, deploying multi-tier water quality classification, and scaling PPP-financed water infrastructure.

Water – Phase 2: Acceleration and Scale-up (2029 – 2032)

Proposed Actions and Objectives

Government Actions	Instruments	Safeguards	Private-Public Cooperation
- Enforce scarcity-indexed water tariffs for industrial and agricultural consumers based on basin stress indices - Mandate ZLD compliance for all new industrial facilities above a defined size threshold - Publish multi-tier water quality classification and enforce quality-matching in industrial permits - Award first 3–5 PPP concessions for water infrastructure under standardised model agreement	- Reduced demand in high-stress basins through tariff signals, increased capital allocation toward infrastructure in water-constrained regions, and improved economic efficiency in water allocation - Green Climate Fund / ADB Water Security Investment Programme - Digital water metering mandate for large agricultural water users	- Annual tariff impact assessment published - ZLD hardship provision for SMEs (phased compliance timeline of additional period) - Independent water ombudsman established	- Standardised PPP framework transforms water infrastructure from a state-budget liability into an attractive, bankable asset class - Financial target for PPP-financed water infrastructure commitments by end of Phase 2

Responsible Entities and Enablers

- Ministry of Water Resources
- Ministry of Economy
- Regional Governors
- Private PPP concessionaires
- ADB / World Bank

Water – Phase 2: Acceleration and Scale-up (2029 – 2032)

Risk Register

Risk	Potential Mitigation Measures
Agricultural sector resists water tariff reform	- Phased transition for agricultural users (5-year grace on full tariff implementation) - Irrigation efficiency subsidy programme

PHASE 3: CONSOLIDATION EFFICIENCY:

Among the relevant actions and initiatives for consolidating and integrating water and water management are achieving measurable reduction in water stress indicators, expanding treated wastewater reuse at scale, and closing the loop between industrial water use and agricultural reuse.

Water – Phase 3: Consolidation (2033 – 2038)

Proposed Actions and Objectives

Government Actions	Instruments	Safeguards	Private-Public Cooperation
- Mandate treated wastewater reuse for agriculture in water-stressed basins (minimum quality Tier 3 treated) - Extend ZLD compliance to all industrial facilities above revised (lower) threshold - Publish national water efficiency league table by sector and region - Integrate water data into national circular reporting framework	- Tiered water pricing now fully operational - Multi-tier water quality classification enabling industrial-agricultural water cascading - Technology Innovation Fund for domestic ZLD technology development	- Health-based water quality standards enforced by State Sanitary Inspectorate - Agricultural users trained on treated wastewater quality parameters - Annual basin-level water stress report published	PPP concessions generating operational track records and attracting refinancing

Responsible Entities and Enablers

- Ministry of Water Resources
- Ministry of Agriculture
- Regional Water Utilities
- Private PPP concessionaires

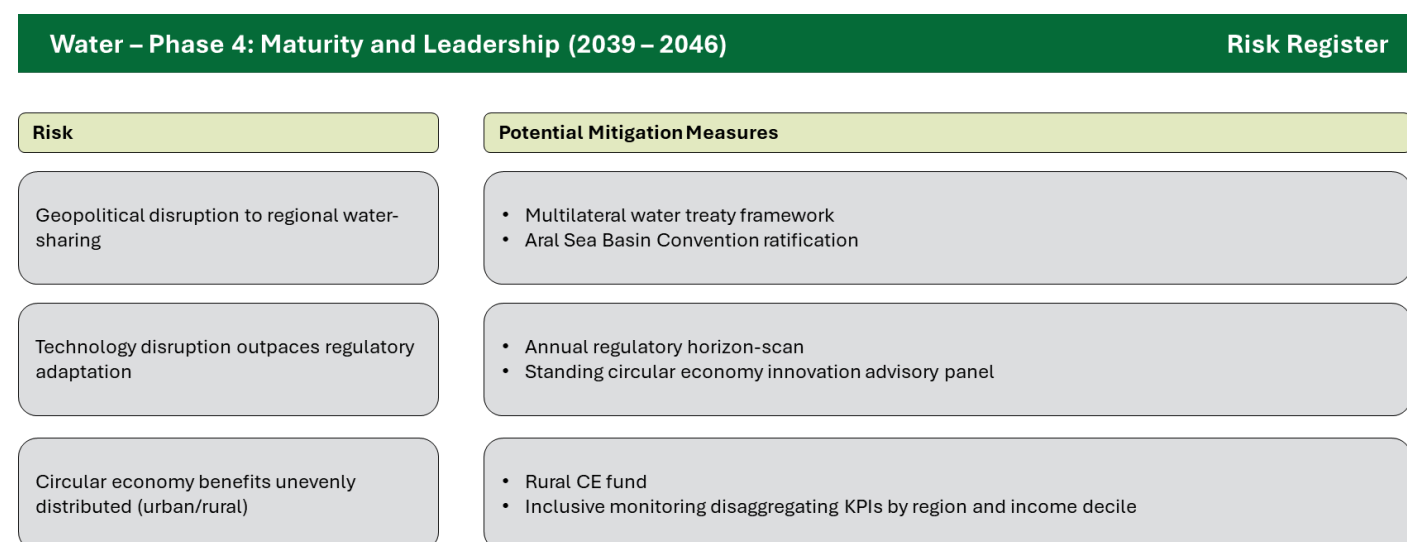
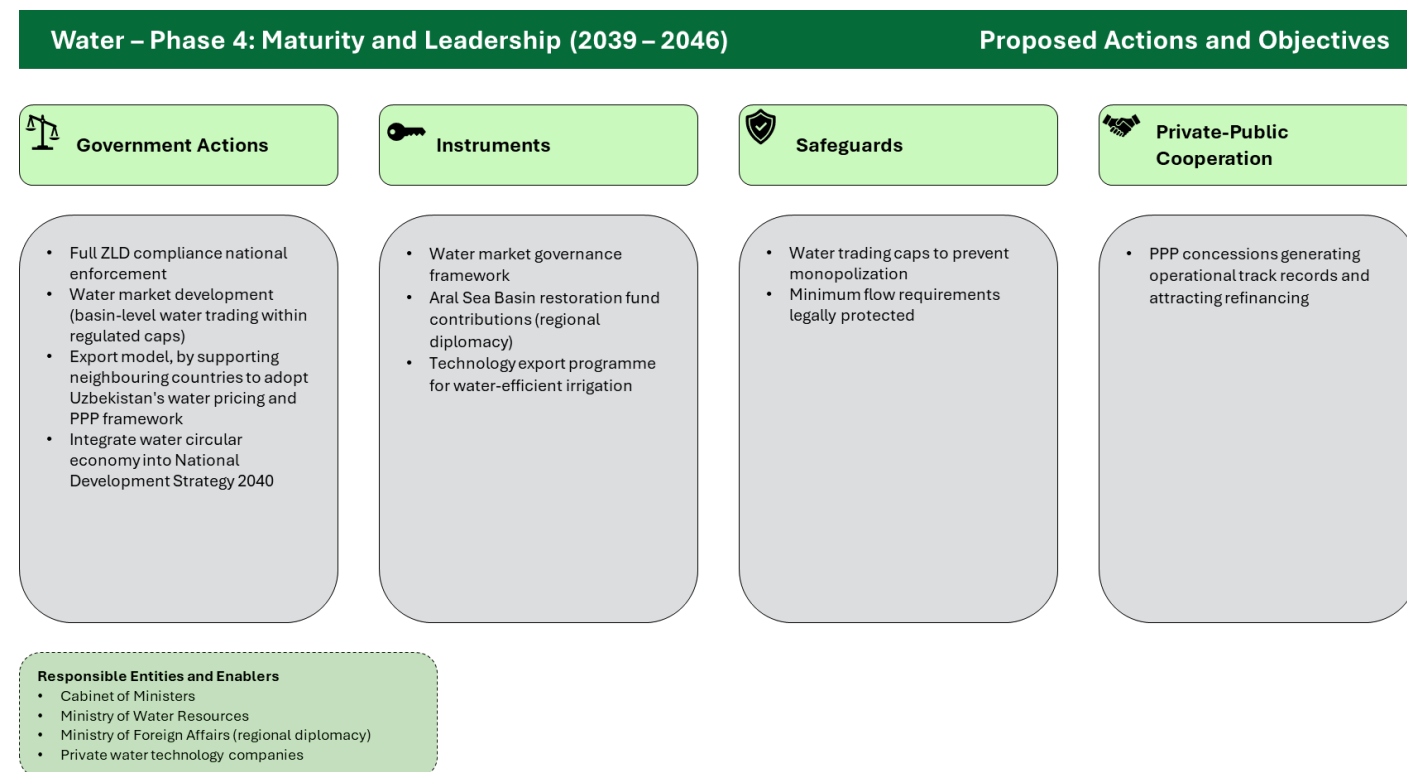
Water – Phase 3: Consolidation (2033 – 2038)

Risk Register

Risk	Potential Mitigation Measures
Climate change accelerates water scarcity faster than infrastructure response	- Accelerate Aral Sea Basin restoration programmes - Activate emergency water-sharing protocols with upstream neighbouring countries

PHASE 4: MATURITY & LEADERSHIP

In this phase, Uzbekistan should assume regional leadership following the development in water efficiency and resource use the past 20 years. Potential objectives are achieving water stress reduction to below 50% in all major basins, expanding treated wastewater reuse to more than 50% of industrial and agricultural demand in water-stressed regions, and achieving achieve full national ZLD compliance for industry.



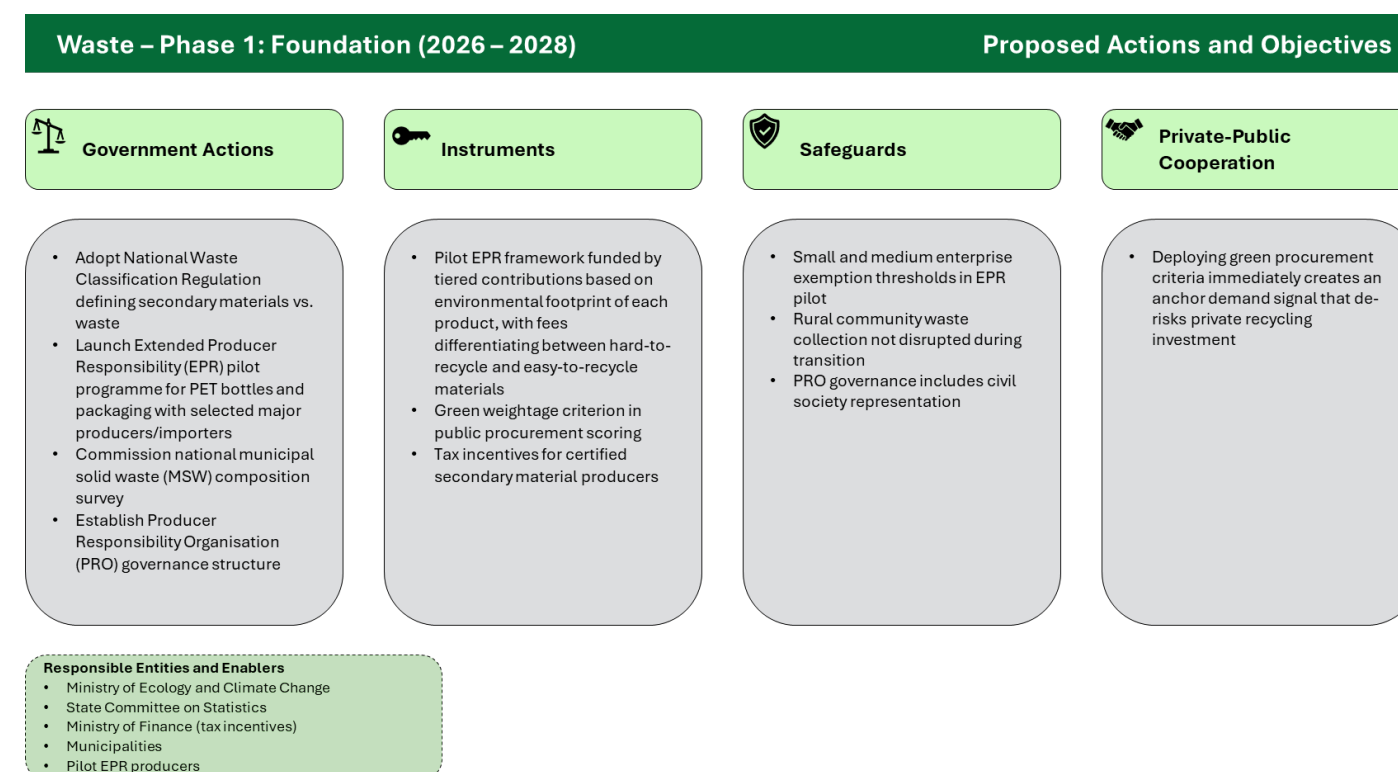
Pillar 2: *Waste*

KEY BARRIERS FOR IMPROVING WASTE MANAGEMENT: EFFICIENCY:

- Government authorities fund recycling infrastructure and operations while producers retain profits from products made from virgin materials, creating no incentive for producers to design their products for recyclability or support collection systems.
- Materials that could be reused in new products are classified and discarded as waste because there is no standardised system to identify, certify, and track materials for second-life applications.
- Waste management operations are informal and fragmented across different operators, making it difficult to implement and scale up consistent recycling practices and environmental standards.
- Low-value plastics persist in the environment for 400+ years and can cause harm to both infrastructure and the environment. The products are sold in low-value, non-recyclable plastics because they are cheap, even though sustainable alternatives are increasingly available, albeit more expensive.

PHASE 1: FOUNDATION

Among the relevant actions and initiatives for building the structural foundation for water efficiency in Uzbekistan are establishing credible water pricing signals, launching industrial zero liquid discharge (ZLD) pilots, and operationalising the October 2025 Water Code through implementing regulations.



Waste – Phase 1: Foundation (2026 – 2028)

Risk Register

Risk	Potential Mitigation Measures
EPR pilot poorly designed, creating perverse incentives	- International benchmarking (EU, Korea EPR models) - Third-party pilot evaluation near end of Phase 1
Low public awareness undermining waste sorting	Immediate national awareness campaign funded by Producer Responsibility Organisation contribution

Waste – Phase 1: Foundation (2026 – 2028)

Potential Barrier Removal Actions

\$ No Producer Incentives for Recyclability	🛒 No Secondary Materials Market
🔗 Fragmented, Informal Operators	🔍 No Waste Composition Data

Removal Mechanisms

Financial liability for post-consumer waste shifts to producers for priority categories	- Second-Life Material Certification standard published - Government procurement preference for certified materials, with bid advantages for “green” solutions
Mandate a single, digital reporting standard for waste flows, replacing unofficial estimates with verified data	Nationally representative municipal solid waste composition survey

PHASE 2: ACCELERATION & SCALE-UP

Among the relevant actions and initiatives for accelerating the improving waste management and use of circular materials are expanding EPR to all plastics nationally, enforcing circular materials preference in all public procurement above a set threshold, launching single-use plastics restrictions, and attracting private investment in sorting/recycling infrastructure.

Waste – Phase 2: Acceleration and Scale-up (2029 – 2032)

Proposed Actions and Objectives

Government Actions	Instruments	Safeguards	Private-Public Cooperation
- Scale EPR from pilot to national scheme covering all plastic packaging and electronics - Introduce single-use plastic restrictions for identified polymer types with a reasonable industry transition period - Mandate circular materials preference in all public contracts above a set threshold - Launch Waste-to-Energy (WtE) concession programme (building on \$1.3B government commitment)	- Multi-stakeholder dialogue to identify specific polymers and applications most detrimental to local environment yet with viable domestic substitutes, allowing transitional period to prevent supply chain disruptions - EPR fees reinvested into PRO-managed collection infrastructure - Green bonds for recycling infrastructure backed by EPR revenue stream	- Informal waste picker integration programme, with an emphasis of formalising livelihoods before collection systems mechanise - Rural collection infrastructure co-funded centrally before EPR fees kick in for rural areas	- Classification system that removes the regulatory ambiguity that prevents companies from reintegrating waste back into the production cycle - Secondary materials market certification providing bankable revenue stream for private recycling operators
Responsible Entities and Enablers - Ministry of Ecology and Climate Change - Ministry of Economy (procurement) - Ministry of Finance (green bonds) - PRO (EPR administration) - Municipal authorities			

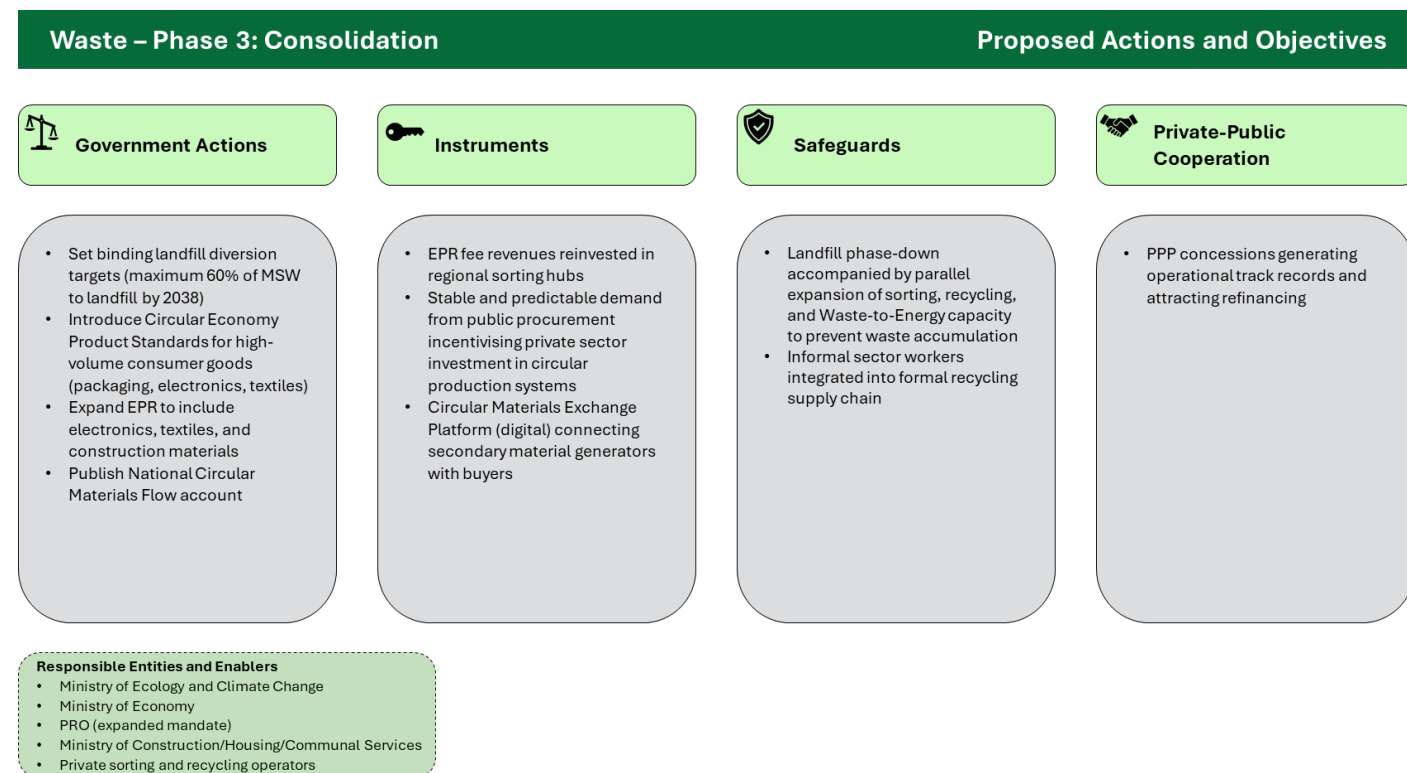
Waste – Phase 2: Acceleration and Scale-up (2029 – 2032)

Risk Register

Risk	Potential Mitigation Measures
Secondary materials market does not develop organically	- Anchor demand through public procurement mandate - Government-backed secondary materials price floor throughout period
Waste-to-Energy projects attract lower-quality operators	- Standardised PPP framework with performance bonds and technical specifications - Independent environmental compliance audit

PHASE 3: CONSOLIDATION

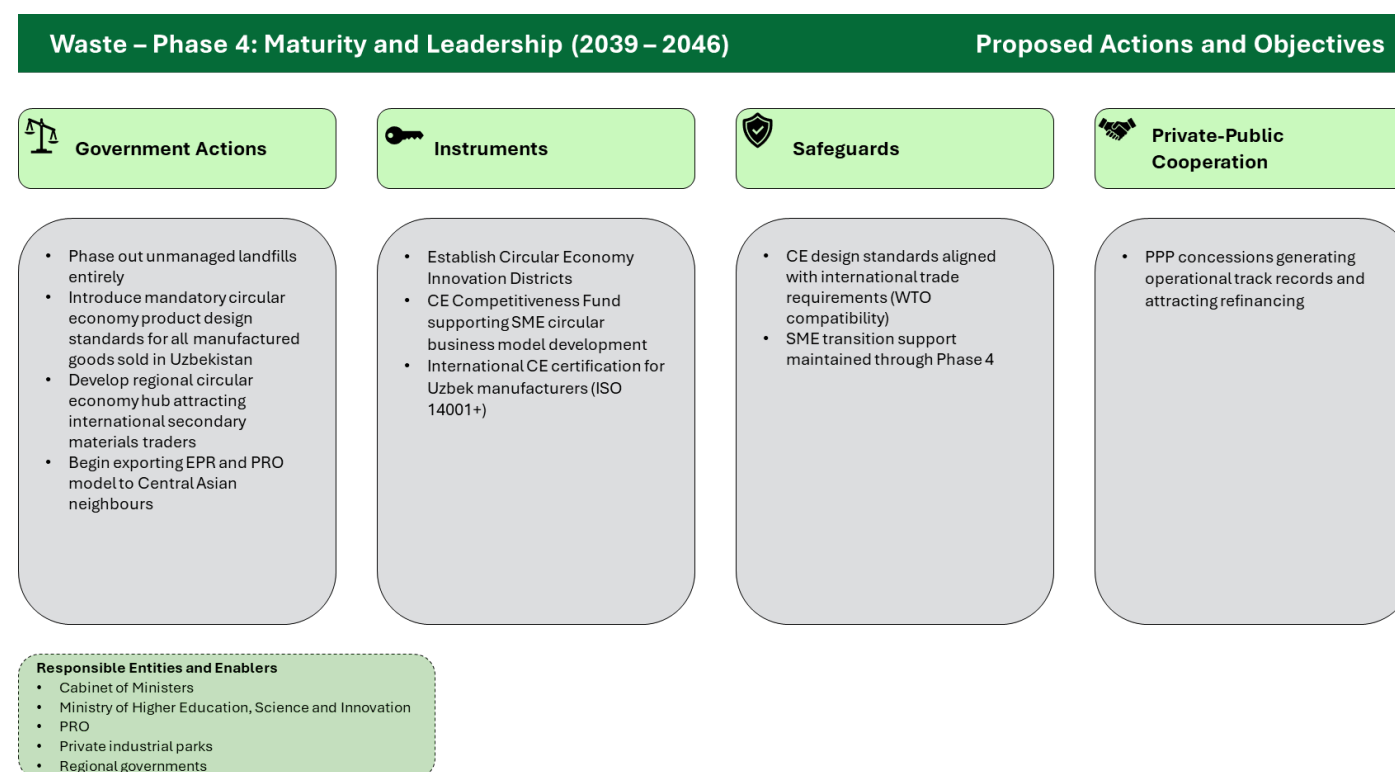
Among the relevant actions and initiatives for consolidating and integrating waste management and best practice are achieving a national recycling rate for municipal waste above 30%, operationalising circular materials economy, and beginning landfill phase-down.



Waste – Phase 3: Consolidation		Risk Register
Risk	Potential Mitigation Measures	
Recycling infrastructure regional gaps leave rural areas excluded	- Central co-funding for rural collection - Mobile sorting units - Regional hub consolidation model	

PHASE 4: MATURITY & LEADERSHIP

In this phase, Uzbekistan should assume regional leadership following the development in waste improvements the past 20 years. Potential objectives are achieving national municipal waste recycling rate above 40%, establishing a circular material use rate (CMUR) above 10%, and achieving near-zero landfilling of recyclable and organic waste.



Waste – Phase 4: Maturity and Leadership (2039 – 2046)		Risk Register
Risk	Potential Mitigation Measures	
Technology disruption outpaces regulatory adaptation	- Annual regulatory horizon-scan - Standing circular economy innovation advisory panel	
Circular economy benefits unevenly distributed (urban/rural)	- Rural CE fund - Inclusive monitoring disaggregating KPIs by region and income decile	

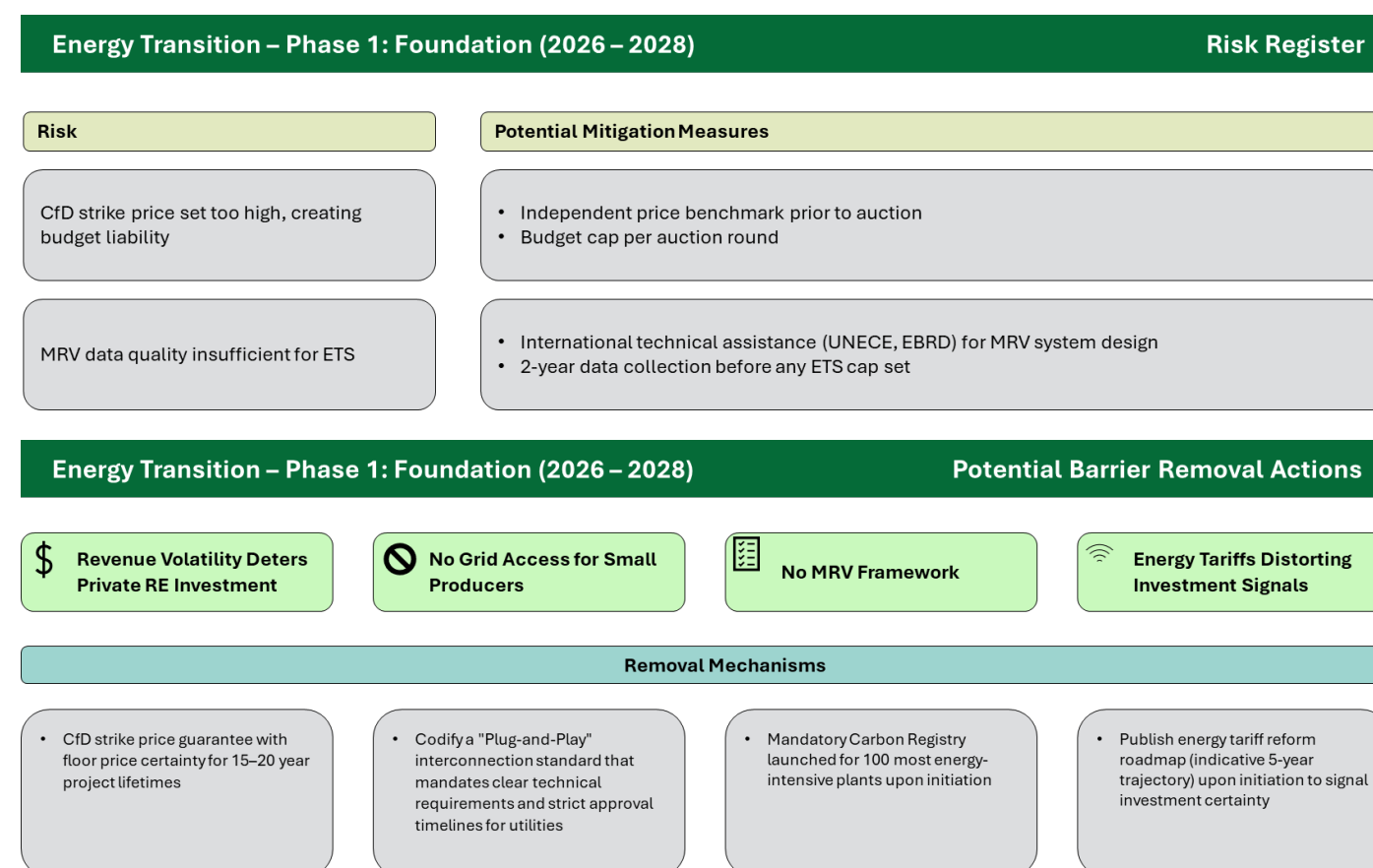
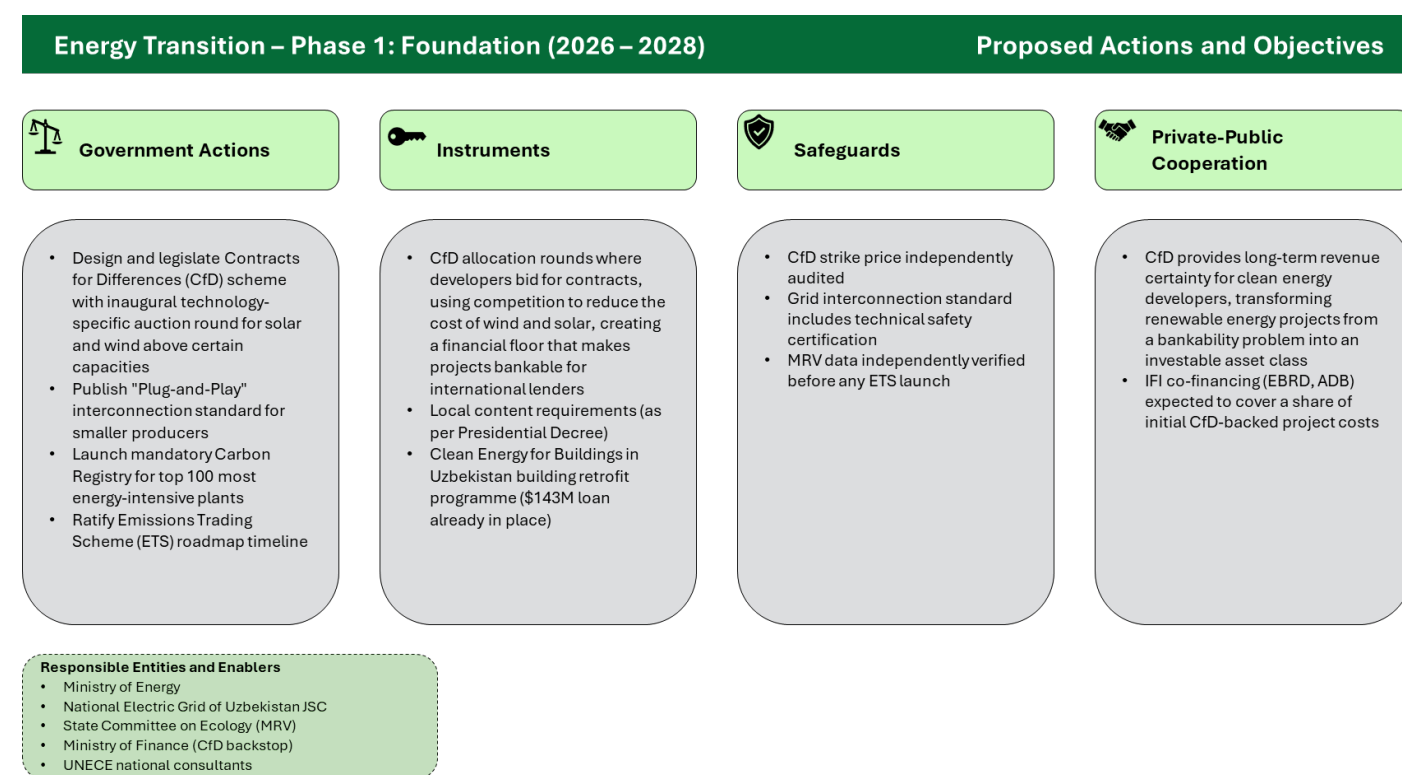
Pillar 3: *Energy Transition*

KEY BARRIERS FOR PREPARING THE ENERGY TRANSITION:

- Renewable energy projects require massive upfront capital but face 25 years of unpredictable power prices, which can crash during peak production. Without a Contracts for Difference (CfD) mechanism, commercial banks view this price volatility as a risk and can refuse to provide the financing required for utility-scale deployment.
- Small-scale renewable energy producers and off-grid solutions lack clear, standardised procedures to connect to the national grid to sell any surplus power, limiting overall renewable energy deployment.
- The most important missing piece is a standardised framework for Measurement, Reporting, and Verification (MRV) of greenhouse gas emissions, leaving the government without the granular data needed to manage national decarbonisation targets. Without an established carbon price, there is no financial incentive for factories to modernise, resulting in continued reliance on inefficient, high-emission technologies.
- As global markets implement carbon border taxes (like the EU's CBAM), local exporters face the threat of losing competitiveness because they cannot provide verified emissions data.

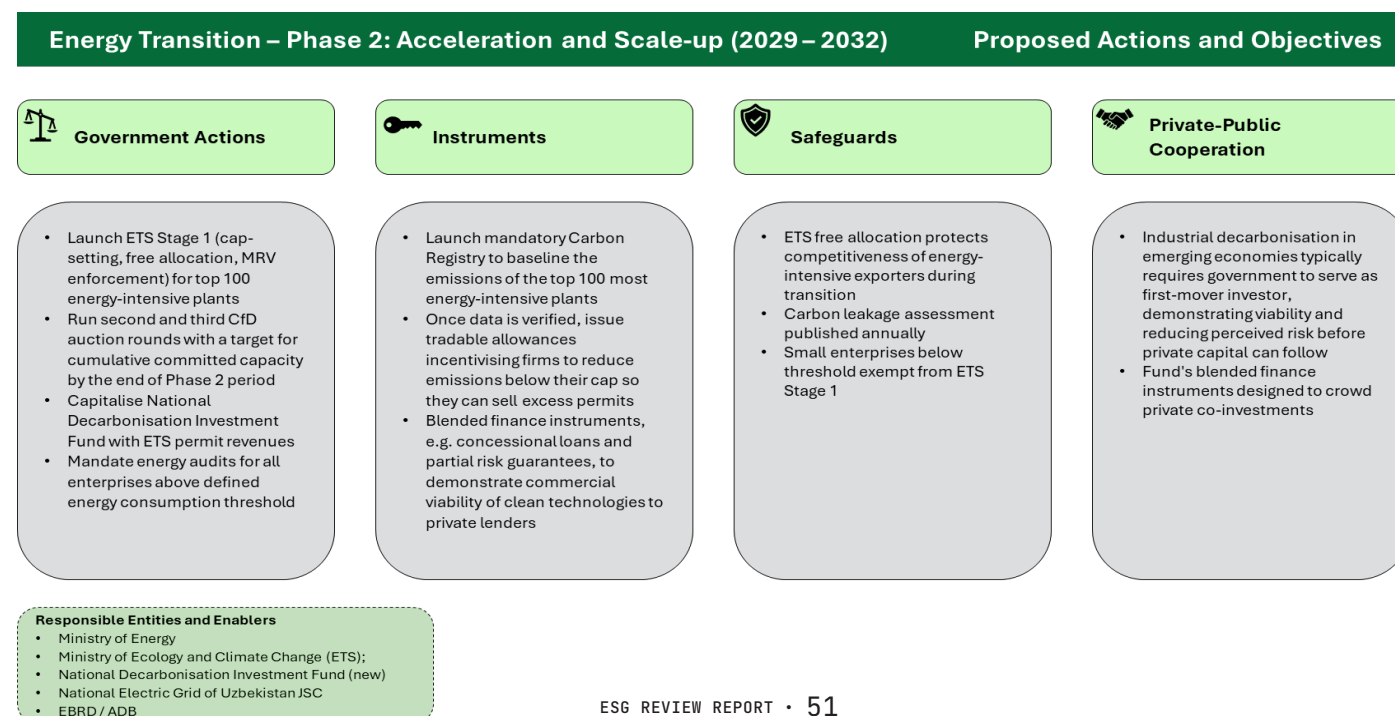
PHASE 1: FOUNDATION

Among the relevant actions and initiatives for building the structural foundation for the energy transition and a low-carbon economy are the launching of a Contracts for Difference (CfD) mechanism design, codifying a Grid Access Standard for small producers, and establishing a Measurement, Reporting, and Verification (MRV) framework as the precursor to the a future Emissions Trading System (ETS).



PHASE 2: ACCELERATION & SCALE-UP

Among the relevant actions and initiatives for accelerating the energy transition are launching domestic Emission Trading Scheme (ETS), scaling CfD-backed renewable capacity, operationalising a National Decarbonisation Investment Fund, and initiating industrial energy efficiency mandates.



Energy Transition – Phase 2: Acceleration and Scale-up (2029 – 2032)

Risk Register

Risk	Potential Mitigation Measures
ETS free allocation set too generously, carbon price too low	- Annual cap review mechanisms - Alignment with Kazakhstan ETS benchmark - Independent verification of emissions data
Grid infrastructure insufficient to absorb rapid RE scale-up	- Accelerate grid modernisation investment as per national strategy) - BESS (battery storage) included in CfD auction design

PHASE 3: CONSOLIDATION

Among the relevant actions and initiatives for further promoting the energy transition and its positive consequences among Uzbek households and in the industry are achieving ~30% renewable share in electricity generation, ensuring that the Emissions Trading Scheme is fully operational with tradable allowances, and accommodating full capitalization of the Decarbonisation Fund and the deployment of industrial decarbonisation finance.

Energy Transition – Phase 3: Consolidation (2033 – 2038)

Proposed Actions and Objectives

Government Actions	Instruments	Safeguards	Private-Public Cooperation
- Enforce ETS Stage 2 (full tradable allowances, annual cap reduction trajectory) - Launch fourth and fifth CfD rounds targeting offshore wind (Aral Sea region) and geothermal - Mandate green building standards for all new public buildings - Introduce Carbon Border Adjustment alignment measures for Uzbek exporters	- Kazakhstan's ETS model provides a regional benchmark for Uzbekistan's Stage 2 ETS cap trajectory - National Decarbonisation Fund deploying concessional loans and partial risk guarantees at scale	- ETS revenues ring-fenced, with a minimum % reinvested in Decarbonisation Fund, and a cap % to household energy affordability support - Just transition plan for gas-dependent workers and communities	- Industrial decarbonisation in emerging economies typically requires government to serve as first-mover investor, demonstrating viability and reducing perceived risk before private capital can follow - Fund's blended finance instruments designed to crowd private co-investments
Responsible Entities and Enablers - Ministry of Energy - Ministry of Ecology and Climate Change (ETS) - National Decarbonisation Investment Fund (new) - Private energy developers - EBRD / ADB / Green Climate Fund			

Energy Transition – Phase 3: Consolidation (2033 – 2038)

Risk Register

Risk	Potential Mitigation Measures
ETS carbon price insufficient to drive industrial fuel-switching	- Floor price mechanism - Review and ratchet cap trajectory at Phase 3 midpoint (2035)
Gas sector lobbying delays tariff reforms and CfD competitiveness	Independent energy regulator established by 2033 with statutory mandate insulated from ministerial interference

PHASE 4: MATURITY & LEADERSHIP

In this phase, Uzbekistan should assume regional leadership following the energy transition the past 20 years. Potential objectives are achieving ~40%+ renewable share in total final energy consumption, reducing energy intensity to within the OECD average for peer economies, and achieving GHG emissions per capita on a Paris Agreement-aligned trajectory.

Energy Transition – Phase 4: Maturity and Leadership (2039 – 2046)

Proposed Actions and Objectives

Government Actions	Instruments	Safeguards	Private-Public Cooperation
- Full ETS Stage 3 (full auctioning, no free allocation for large emitters) - Green hydrogen development strategy launched - Uzbekistan joins international carbon market linkage (Kazakhstan ETS bridge) - Mandatory carbon disclosure for all listed companies	- Mature National Decarbonisation Fund - Green bond market established - Sovereign green bond issuance - International carbon market linkage revenues	- Energy affordability monitoring to prevent energy poverty - Just transition completed for gas-dependent communities	- Industrial decarbonisation in emerging economies typically requires government to serve as first-mover investor, demonstrating viability and reducing perceived risk before private capital can follow - Fund's blended finance instruments designed to crowd private co-investments
Responsible Entities and Enablers - Ministry of Energy - Ministry of Finance - National Decarbonisation Fund (new) - Tashkent Stock Exchange (green bonds) - Private energy sector			

Energy Transition – Phase 4: Maturity and Leadership (2039 – 2046)

Risk Register

Risk	Potential Mitigation Measures
Technology disruption outpaces regulatory adaptation	- Annual regulatory horizon-scan - Standing circular economy innovation advisory panel
Circular economy benefits unevenly distributed (urban/rural)	- Rural CE fund - Inclusive monitoring disaggregating KPIs by region and income decile
Global carbon market price volatility undermining ETS	Price stability corridor mechanism in ETS legislation

Appendix

APPENDIX 1 – GLOSSARY

Circular Economy (CE)

A regenerative system designed to minimize resource input, waste, emissions, and energy leakage by slowing, closing, and narrowing material and energy loops. Unlike the traditional linear economy, which follows a “take-make-dispose” pattern, a circular model prioritizes longer-lasting design, maintenance, repair, and remanufacturing.

Linear Economy

The traditional model of “take, make, dispose,” in which resources are extracted and consumed without regard for resource loss or the afterlife of products, as opposed to circular principles that design out waste and keep materials in use for as long as possible.

The 9Rs Framework

Nine essential strategies for transitioning from a linear to a circular economy, presented from most to least efficient: Refuse, Rethink, Reduce, Reuse, Repair, Refurbish, Remanufacture, Repurpose, and Recycle/Recover. These strategies aim to reduce resource use and waste.

Extended Producer Responsibility (EPR)

A regulatory scheme that holds manufacturers and importers financially accountable for the post-consumer lifecycle of their products, including collection and recycling, thereby shifting the financial burden from public authorities to the private sector.

Producer Responsibility Organisation (PRO)

A governance structure established to administer an EPR scheme on behalf of producers, managing collection infrastructure and disbursing EPR fee revenues. In industry-led systems, producers themselves govern the PRO, driving cost efficiency, performance improvement, and investment in downstream infrastructure.

Circular Material Use Rate (CMUR)

A key performance indicator measuring the share of circular (recovered or recycled) materials used in the economy as a proportion of total material consumption.

Secondary Materials

Materials that could be reused in new products but are currently classified and discarded as waste due to the absence of a standardized system to identify, certify, and track them for second-life applications.

Zero Liquid Discharge (ZLD)

An industrial water management approach in which wastewater is treated to the point where no liquid effluent is discharged off-site.

Water Stress

The ratio of freshwater withdrawal to available freshwater resources, expressed as a percentage. A high water stress level indicates that withdrawals significantly exceed sustainable supply.

Water Cascading

The sequential reuse of water across sectors based on quality classification, enabling industrial wastewater treated to a sufficient tier to be reused for agricultural or other purposes, thereby closing the water loop.

Waste-to-Energy (WtE)

A waste management approach that converts non-recyclable waste into electricity and heat through incineration. WtE diverts waste from landfill and can power district heating networks, but does not constitute recycling in the circular economy hierarchy.

Municipal Solid Waste (MSW)

The aggregate of solid household and community waste streams. A national MSW composition survey is a foundational tool for designing effective waste classification and recycling policies.

Blended Finance

A financing approach that combines public or concessional instruments (e.g., concessional loans and partial risk guarantees) with private capital to demonstrate the commercial viability of clean technologies, lower the cost of capital, and attract international investors to projects in emerging markets.

Monitoring, Reporting, and Verification (MRV)

A standardized framework for measuring and independently verifying greenhouse gas emissions data. Without an established MRV system, governments lack the granular data needed to manage national decarbonization targets or operate an emissions trading scheme.

Emissions Trading Scheme (ETS)

A carbon market mechanism that sets a cap on total emissions and allows regulated entities to buy and sell tradable allowances, creating a financial incentive for industrial operators to reduce their emissions below their cap.

Public-Private Partnership (PPP)

A contractual arrangement under which private operators deliver public infrastructure or services. Standardized model concession agreements that pre-define risk, revenue guarantees, and dispute resolution transform resource infrastructure from a state-budget liability into a bankable asset class.



MINISTRY OF INVESTMENT,
INDUSTRY AND TRADE
OF THE REPUBLIC OF UZBEKISTAN



FOREIGN INVESTORS COUNCIL
UNDER THE PRESIDENT OF
THE REPUBLIC OF UZBEKISTAN

Coca-Cola



ESG REVIEW REPORT

The future is *circular*.

It is time to close the loop.

Uzbekistan stands at an inflection point. The decisions made in the next three to five years will determine whether the country captures the full economic, environmental, and strategic value of the circular transition – or allows legacy linear patterns to constrain the growth ambitions of a nation on the rise.

Water, waste, and energy are one system. One strategy can unlock all three. One Presidential directive can set the direction for a generation.